DEFENSE SLIDES (8/18/25)

THE TANGLENOMICON:

TABULATION OF ARBORESCENT TANGLES

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"The Tanglenomicon" name due to Dr. Nicholas Connolly

KNOTS

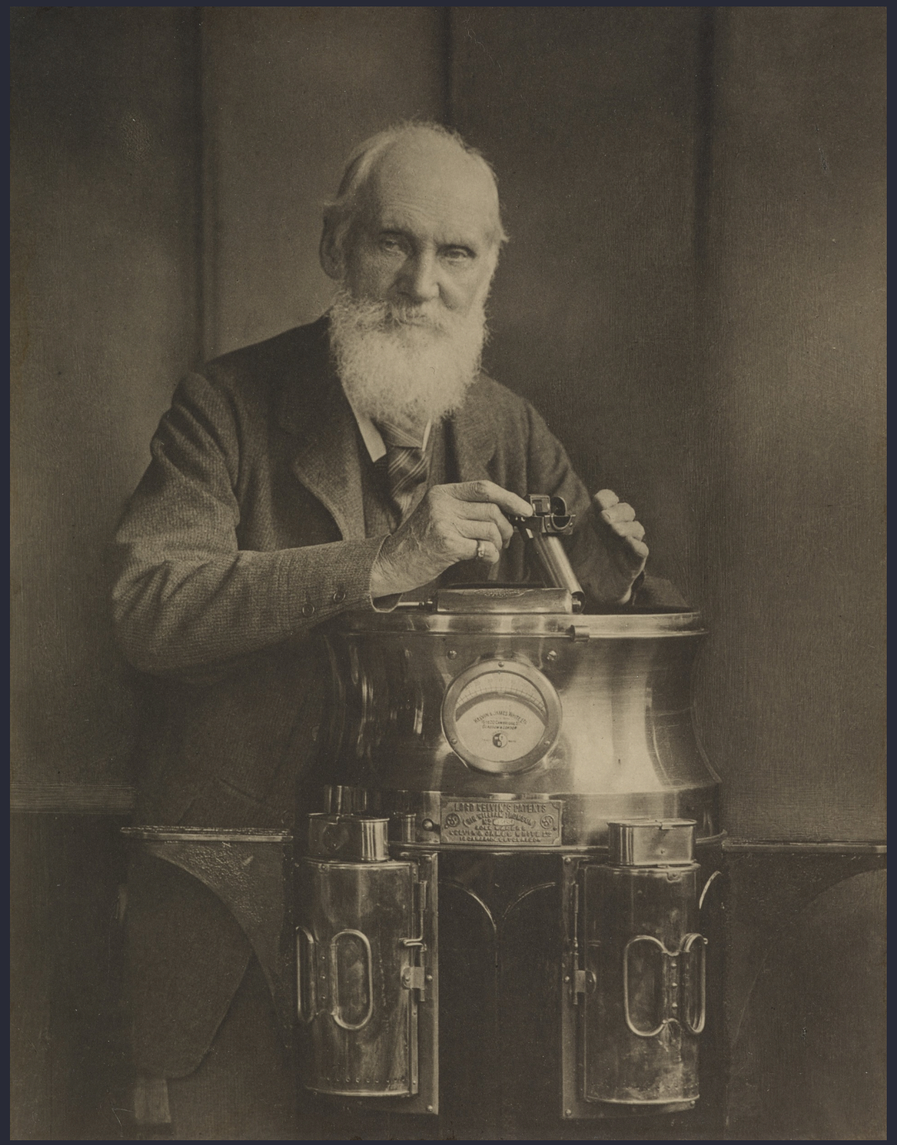
https://www.knotplot.com/

THE NATURAL QUESTION

HOW MANY KNOTS?

KNOT TABLES

LORD KELVIN'S VORTEX THEORY OF THE ATOM



Atoms are knotted vortices in the æther.

BY HAND

- 1860s Tait computes knots up to 7 crossings
    - 15 knots
- 1870s Tait, Kirkman, and Little compute knots up to 10 crossings
    - Takes about 25 years
    - 250 knots (+1 repeat in the Perko pair)
- 1960s Conway computes knots up to 11 crossings
    - “A few hours”
- 1980s Caudron verifies knots up to 11 crossings
    - Finding 4 omissions of Conway

BY COMPUTER

- 1980s Dowker and Thistlethwaite compute up to 13 crossings
  - First using a computer
  - 12,966 knots
- 1990s Hoste, Thistlethwaite, and Weeks compute up to 16 crossings
  - Computer runtime on the order of weeks
  - 1,701,936 knots
- 2020s Burton computes up to 19 crossings
  - Computer runtime on the order of months
  - 350 million knots

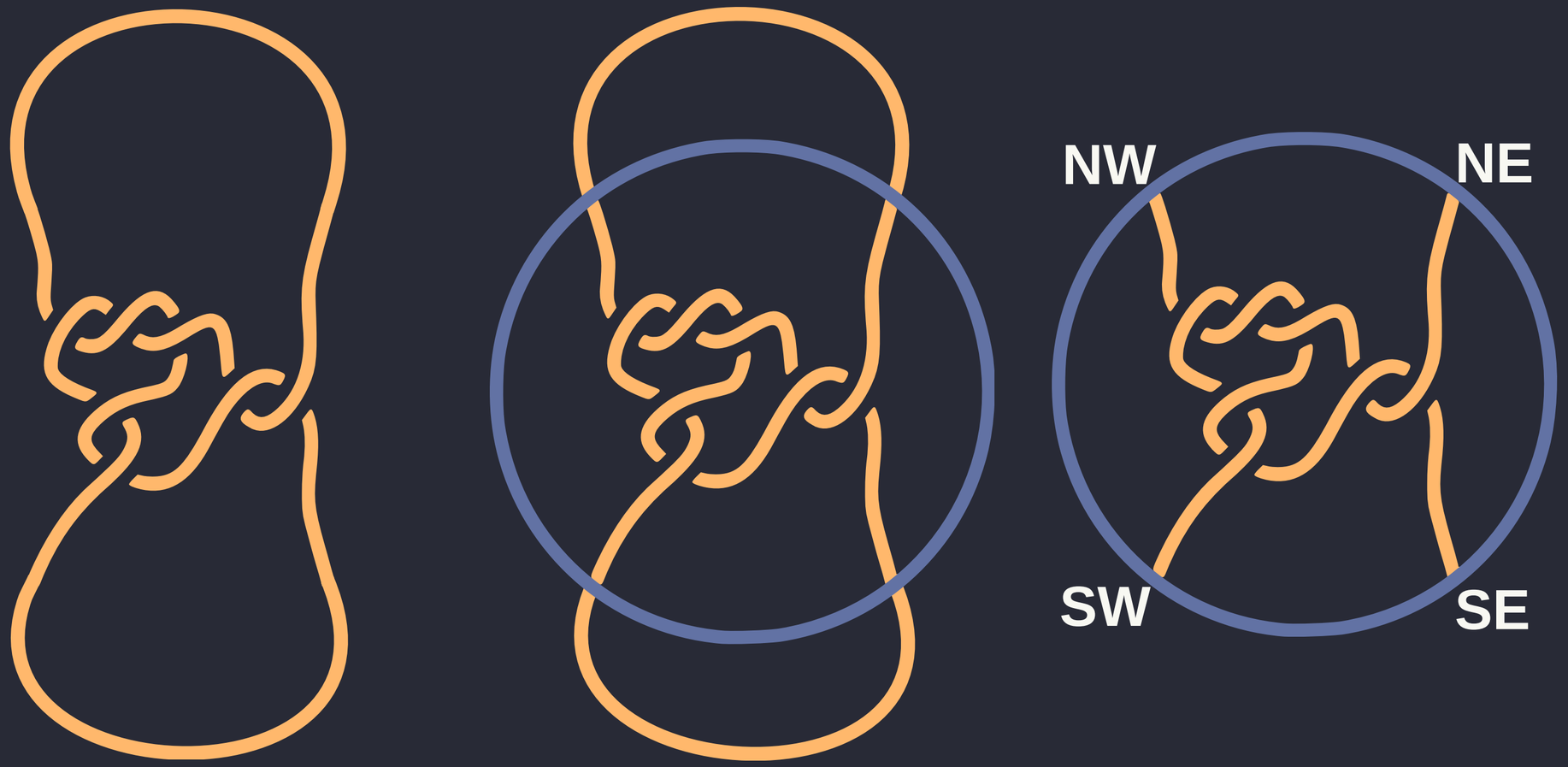
CONWAY

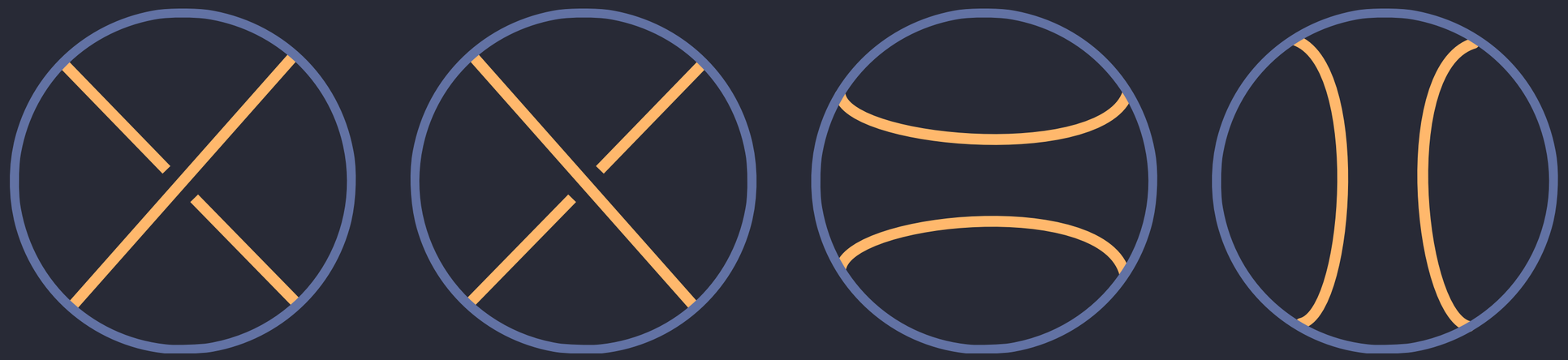
How did Conway compute 25 years of work in "a few hours"?

TANGLES

*"We define a **tangle** as a portion of a knot diagram from which there emerge just 4 arcs pointing in the compass directions NW, NE, SW, SE." - Conway, J.H.*

Conway, J.H. "An Enumeration of Knots and Links, and Some of Their Algebraic Properties." In Computational Problems in Abstract Algebra, 329-58. Elsevier, 1970. https://doi.org/10.1016/B978-0-08-012975-4.50034-5

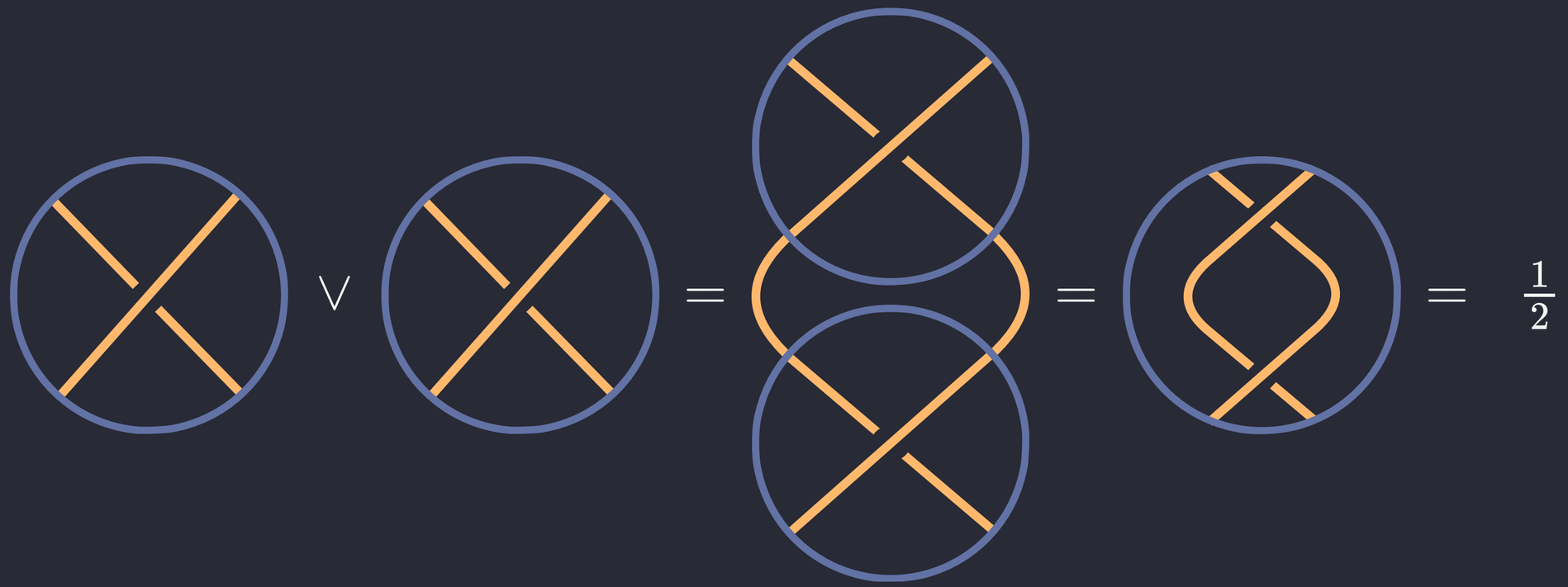

NW
NE
SW
SE



BASIC OPERATIONS

OPERATION $+$

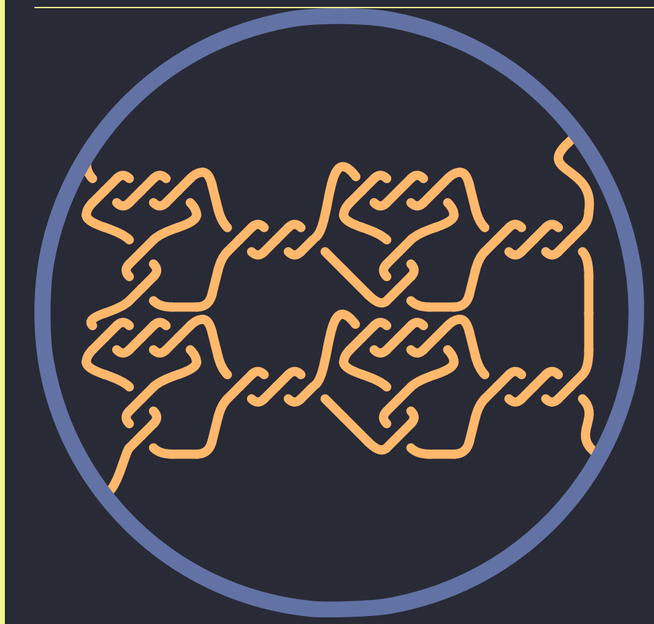
OPERATION $\vee$



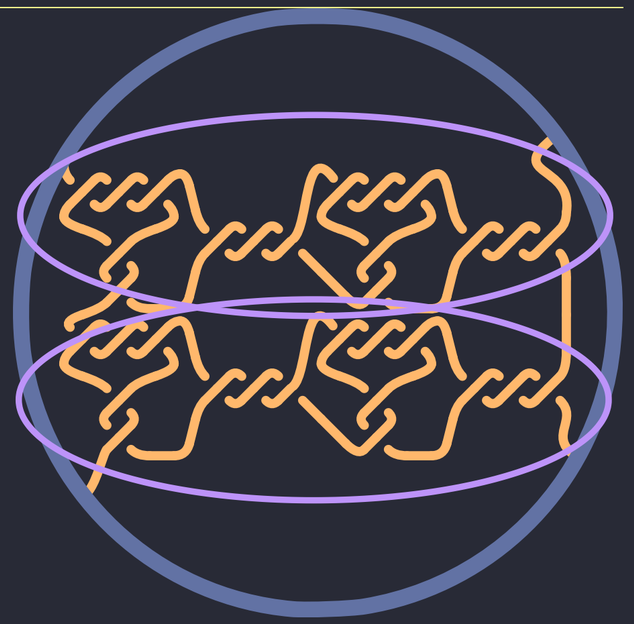
ALGEBRAIC TANGLES

All possible tangles made from $+$ and $\vee$ on basic tangles.

Algebraic



A tangle built from $\vee$ and $+$ on some basic tangles.



$$\left(\left(\left(3 \vee \frac{1}{2}\right) + 3\right) + \left(\left(3 \vee \frac{1}{2}\right) + 3\right)\right) \vee \left(\left(\left(3 \vee \frac{1}{2}\right) + 3\right) + \left(\left(3 \vee \frac{1}{2}\right) + 3\right)\right)$$

$$\vee + + \vee 3 \frac{1}{2} 3 + \vee 3 \frac{1}{2} 3 + + \vee 3 \frac{1}{2} 3 + \vee 3 \frac{1}{2} 3$$

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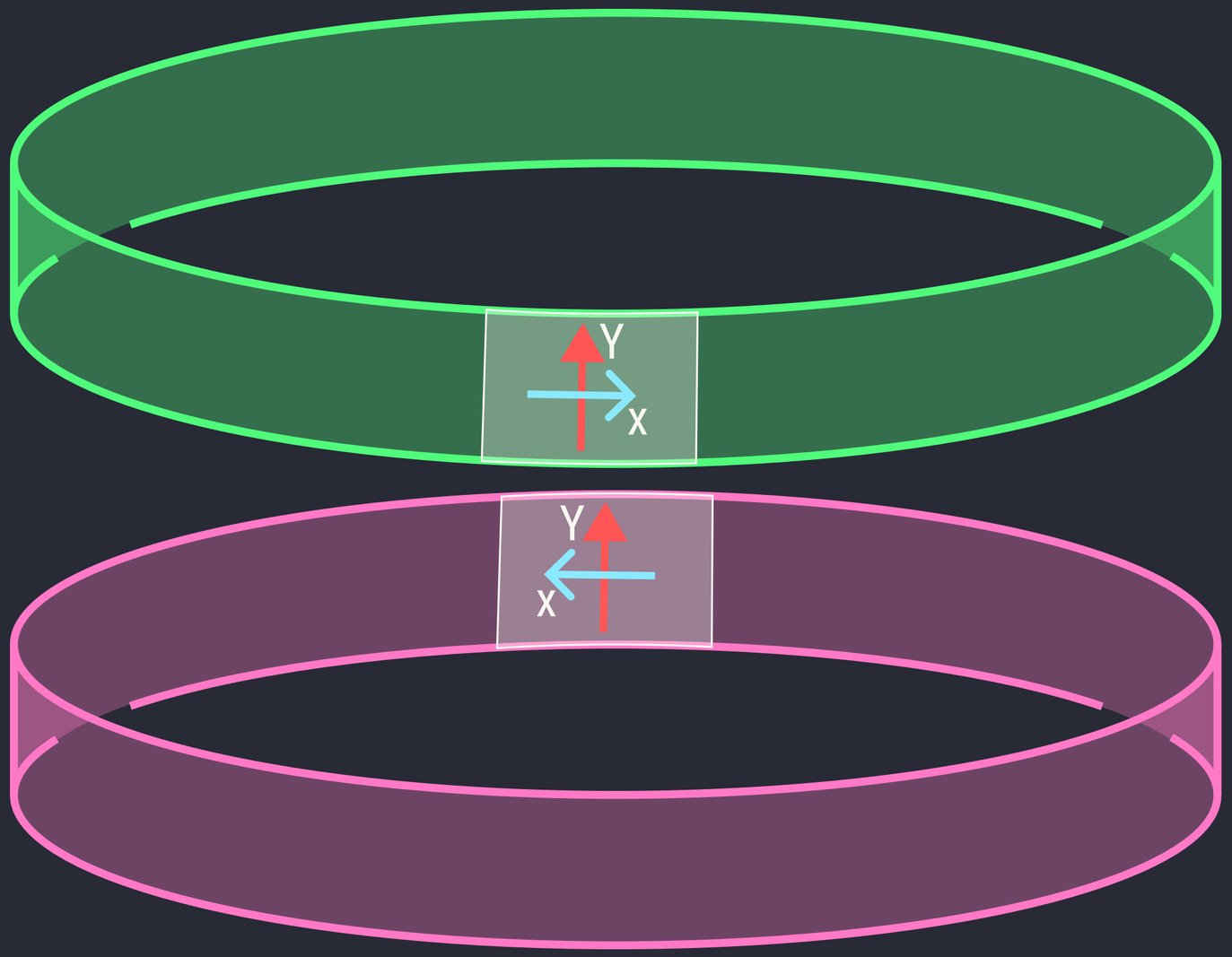
ARBORESCENT TANGLES

Arborescent knots (and tangles) are constructed by taking a collection of twisted bands described by a weighted tree and connecting them with non-cyclic successive plumbing.

Algebraic and arborescent constructions describe the same class of objects.

F. Bonahon and L. Siebenmann, New geometric splittings of classical knots, and the classification and symmetries of arborescent knots, http://www-bcf.usc.edu/~fbonahon/Research/Publications.html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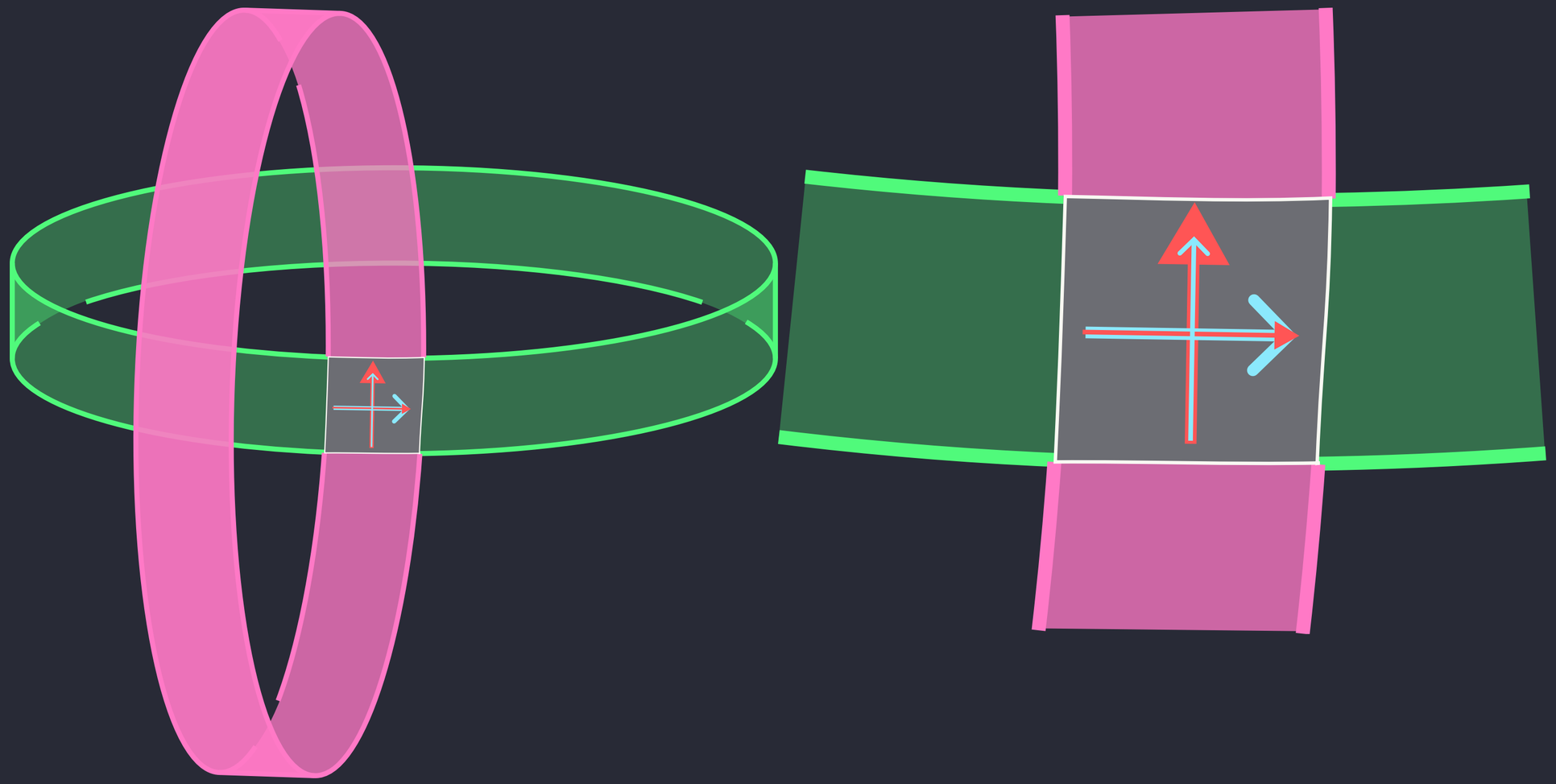
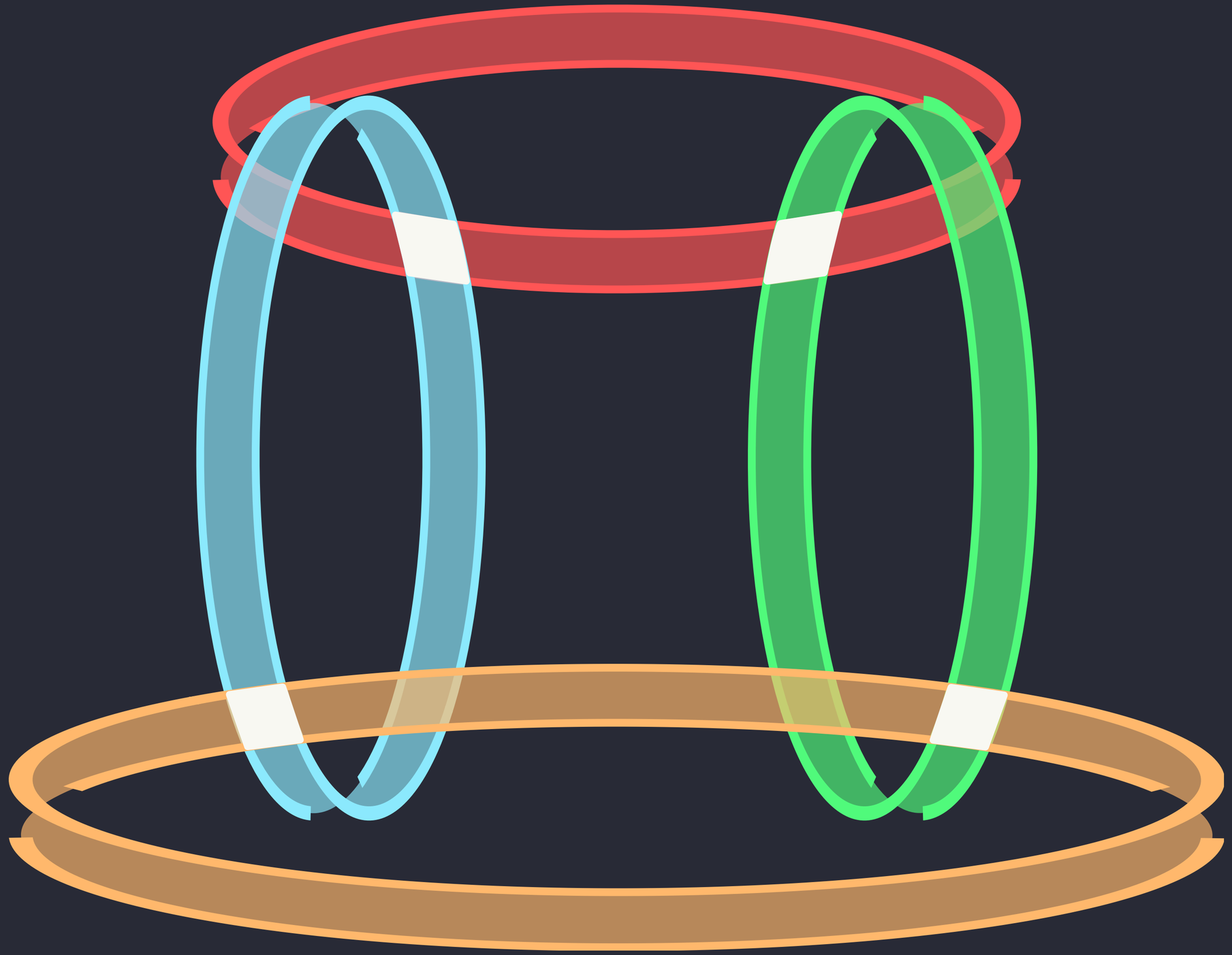
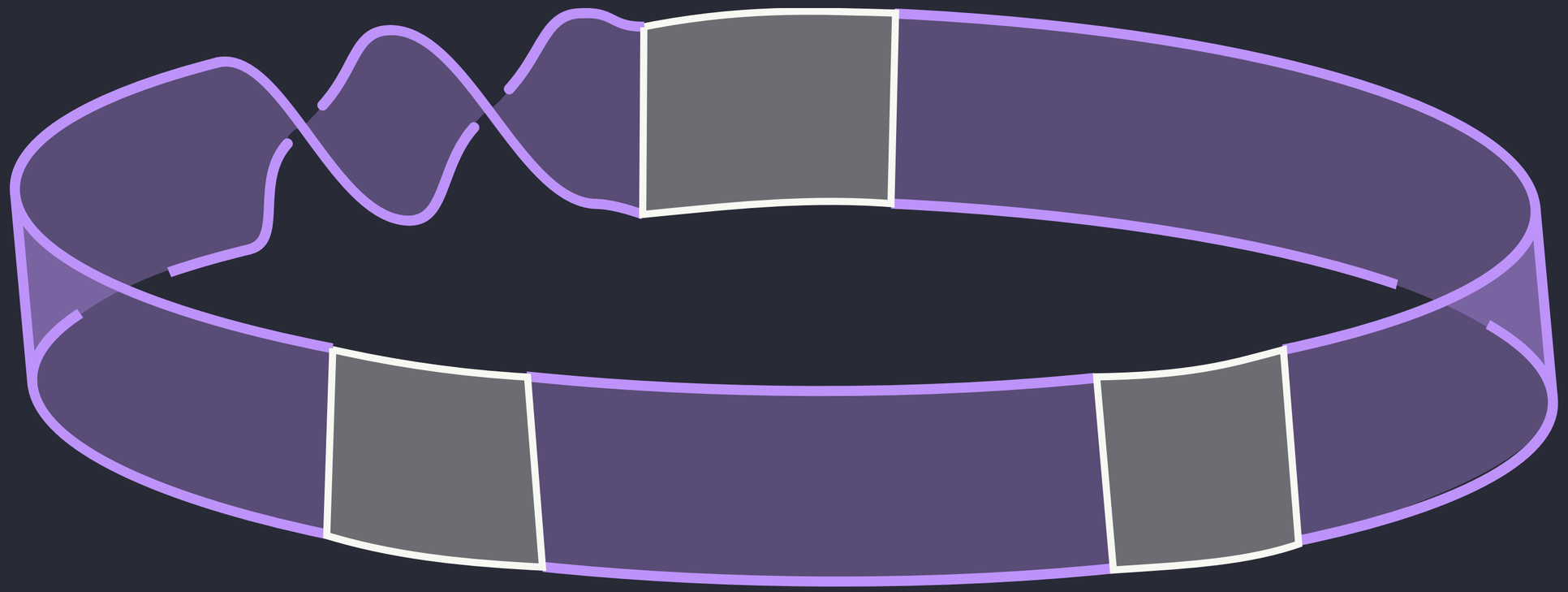
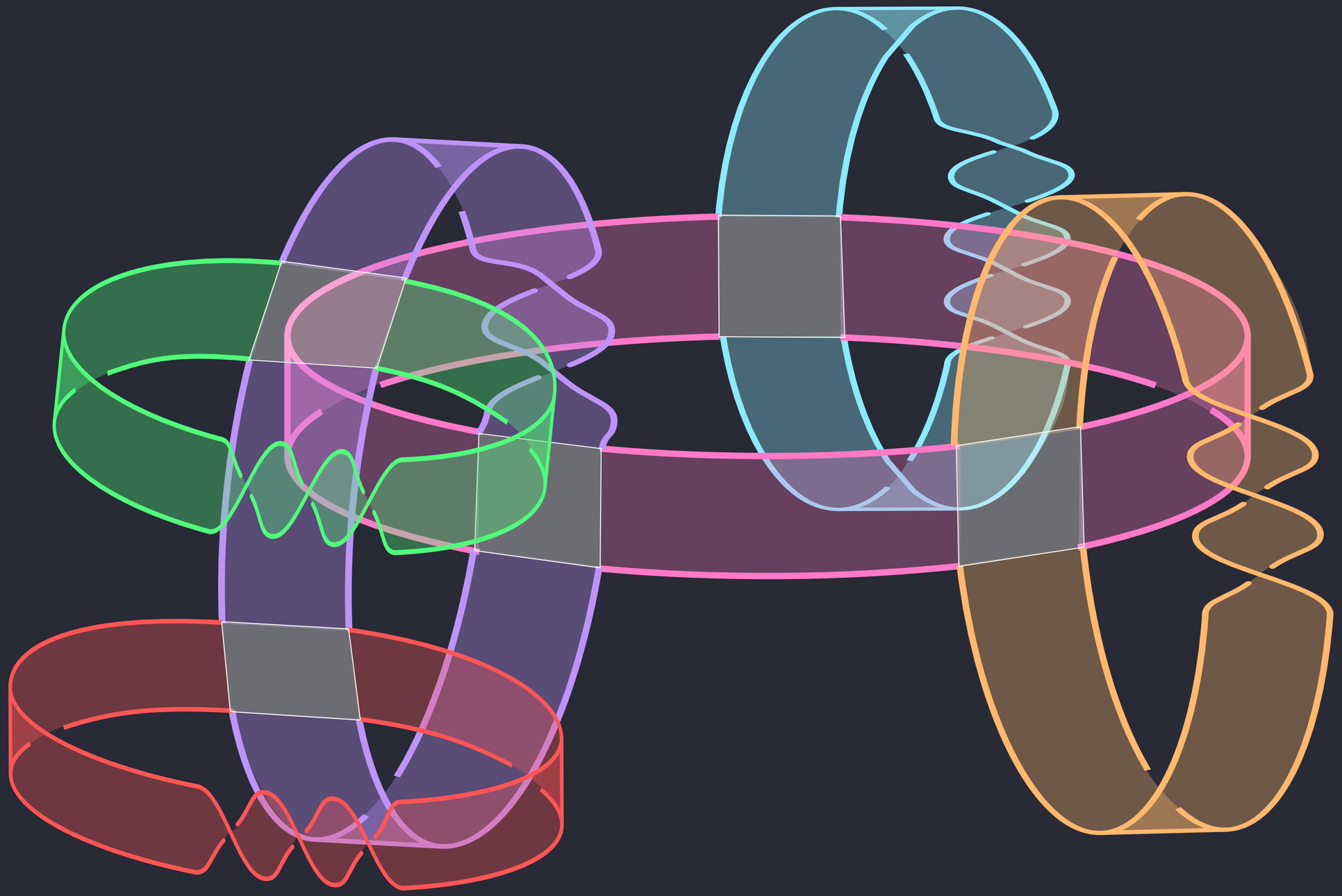
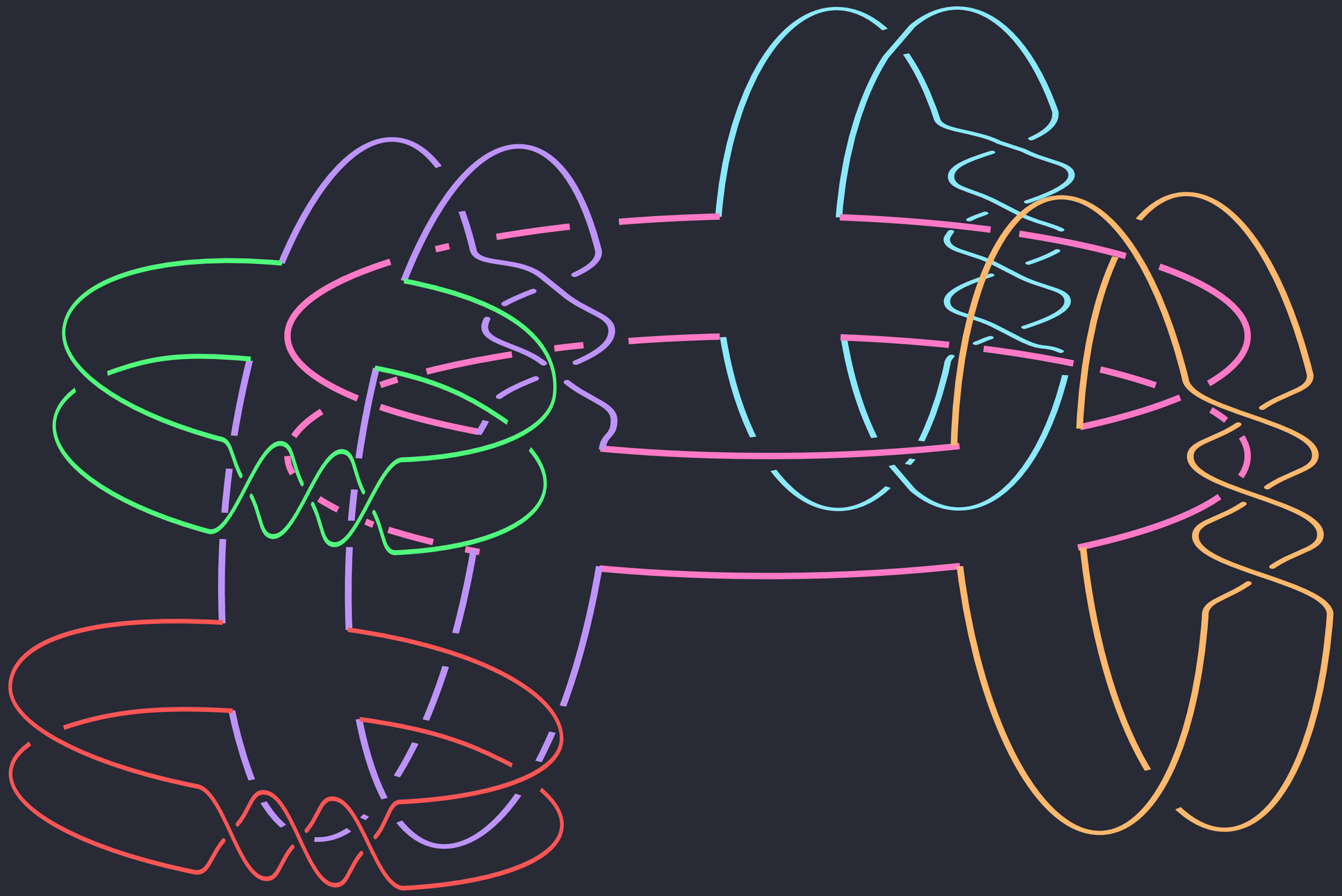
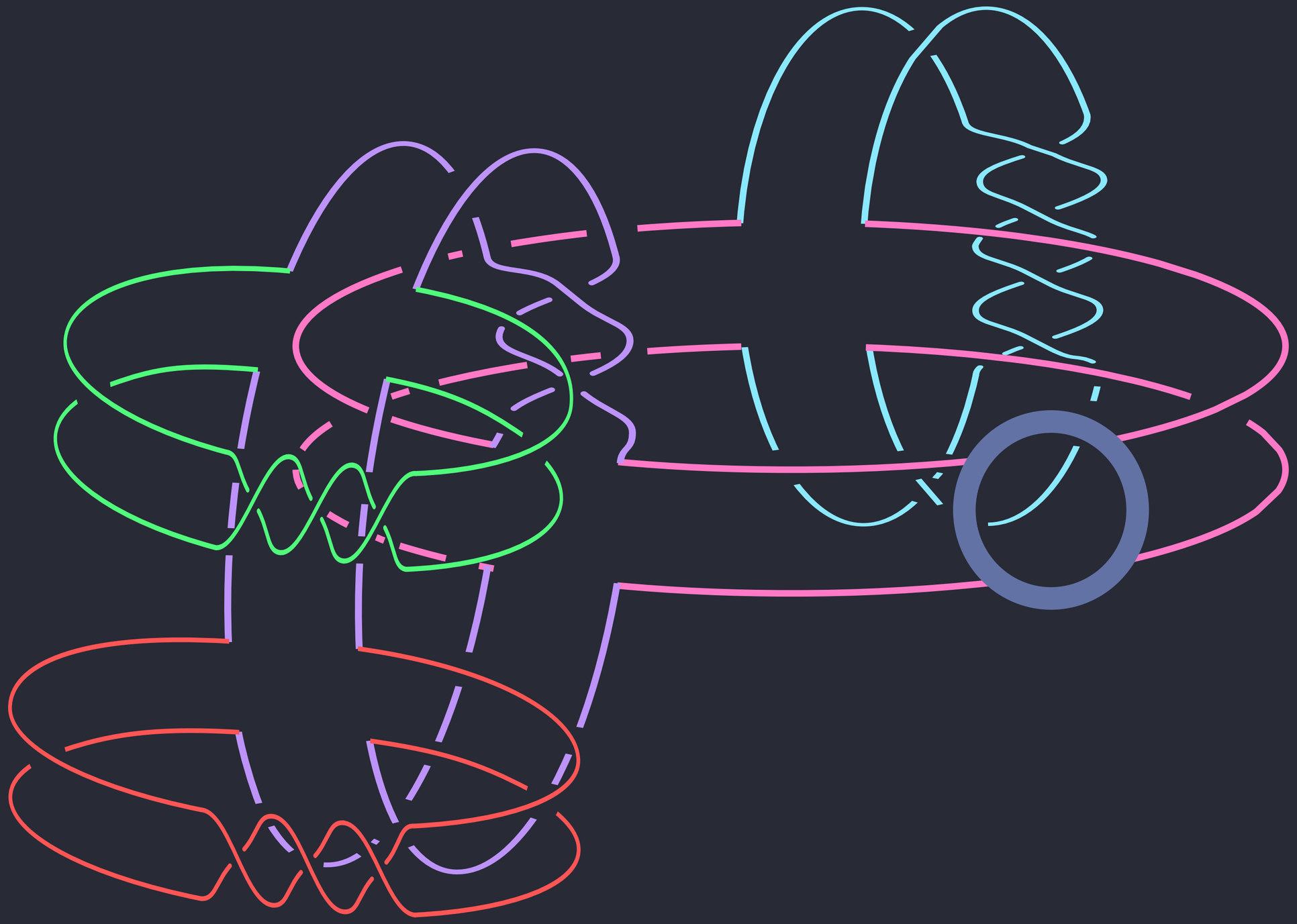


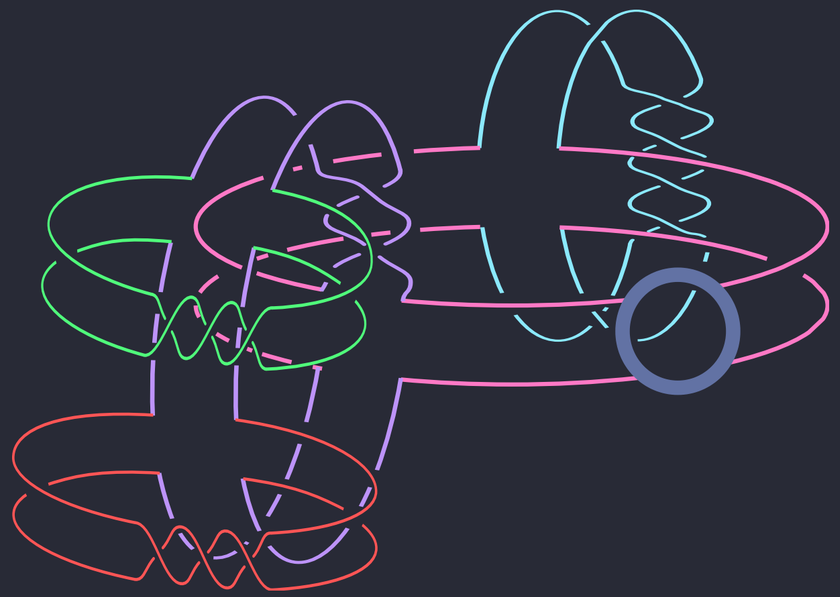












We can see here the correspondence between algebraic and arborescent constructions.

HOW TO ENCODE? (BONAHON AND SIEBENMANN)

1. Relationship between a band and the other bands plumbed to it (children).
2. Location of twists relative to the children.
3. Tangle boundary.

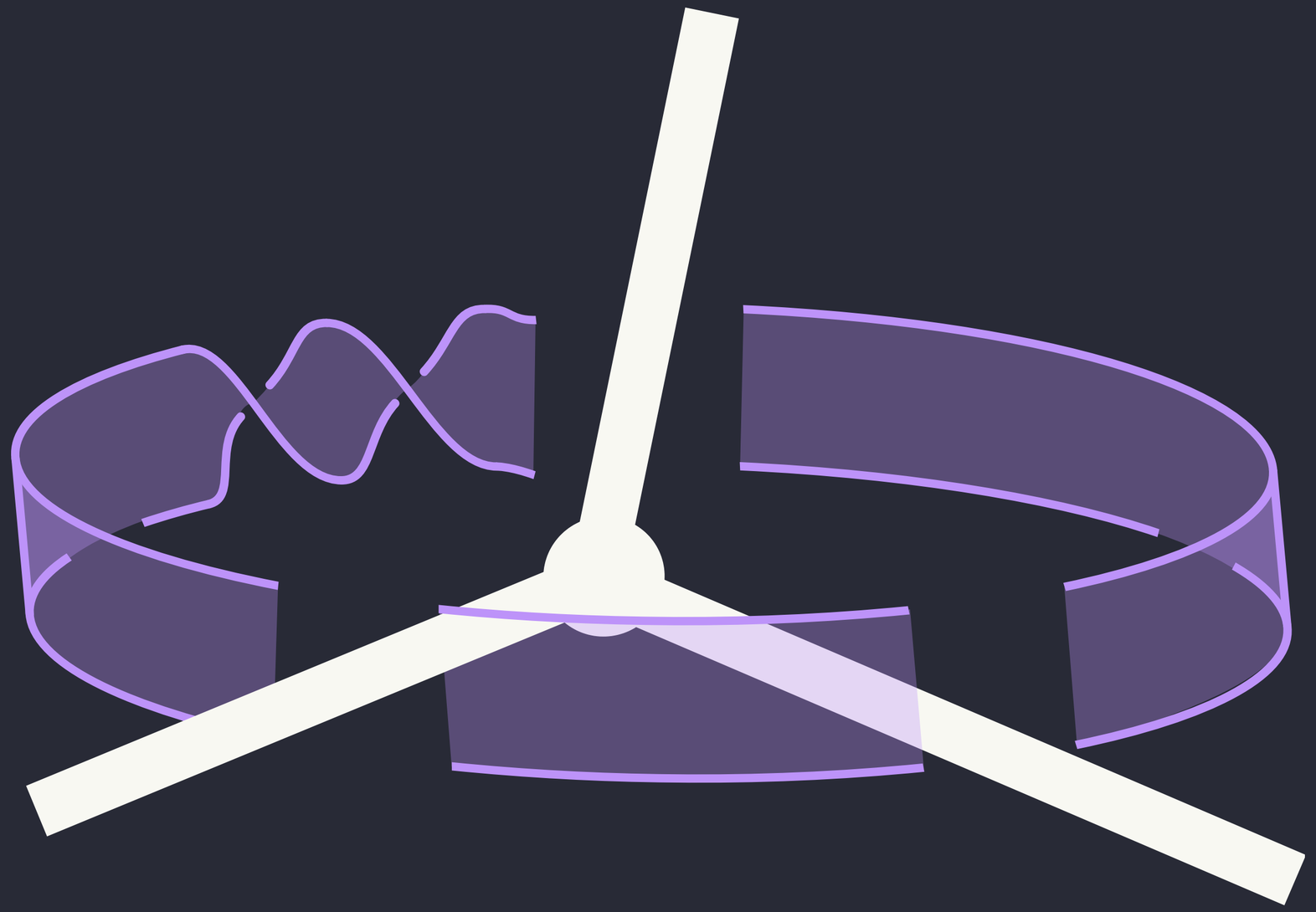
F. Bonahon and L. Siebenmann, New geometric splittings of classical knots, and the classification and symmetries of arborescent knots, http://www-bcf.usc.edu/~fbonahon/Research/Publications.html

1. RELATIONSHIP BETWEEN A BAND AND THE OTHER BANDS ATTACHED TO IT (CHILDREN)

1. Acyclic connections between items.
2. Relative positions of connections.

ACYCLIC CONNECTIONS BETWEEN ITEMS?

SOLUTION: A TREE (IN THE GRAPH SENSE)



RELATIVE POSITIONS OF CONNECTIONS?

SOLUTION: ROOTED PLANE TREE

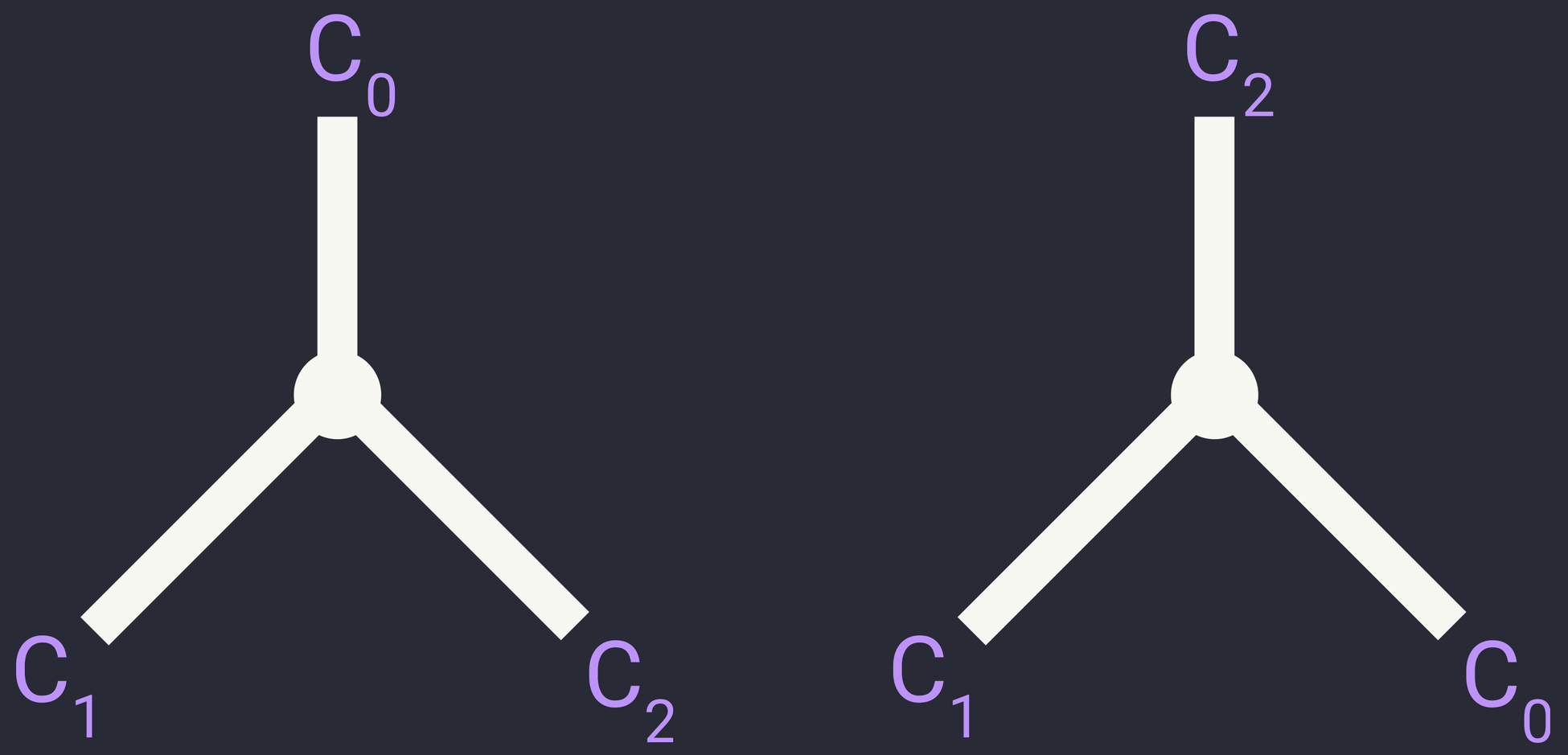
Definition

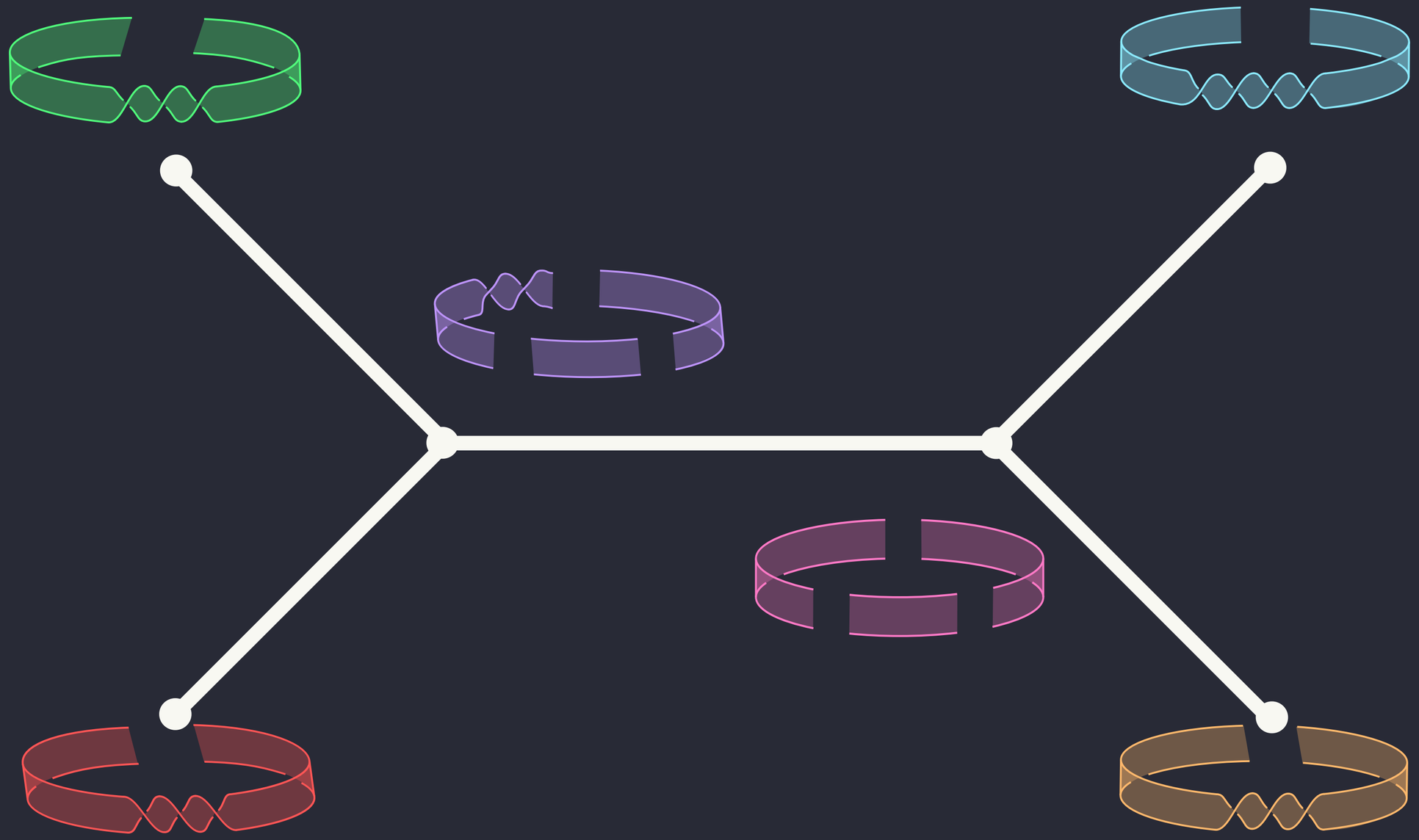
A **rooted plane tree** is an abstract tree imbued with a strict total order on the vertices (indexed by the non-negative integers). We call the least vertex the **root** of the tree.

Convention:

We will select the total order given by a depth first in order traversal of the tree.

CYCLIC ORDER OF A VERTEX





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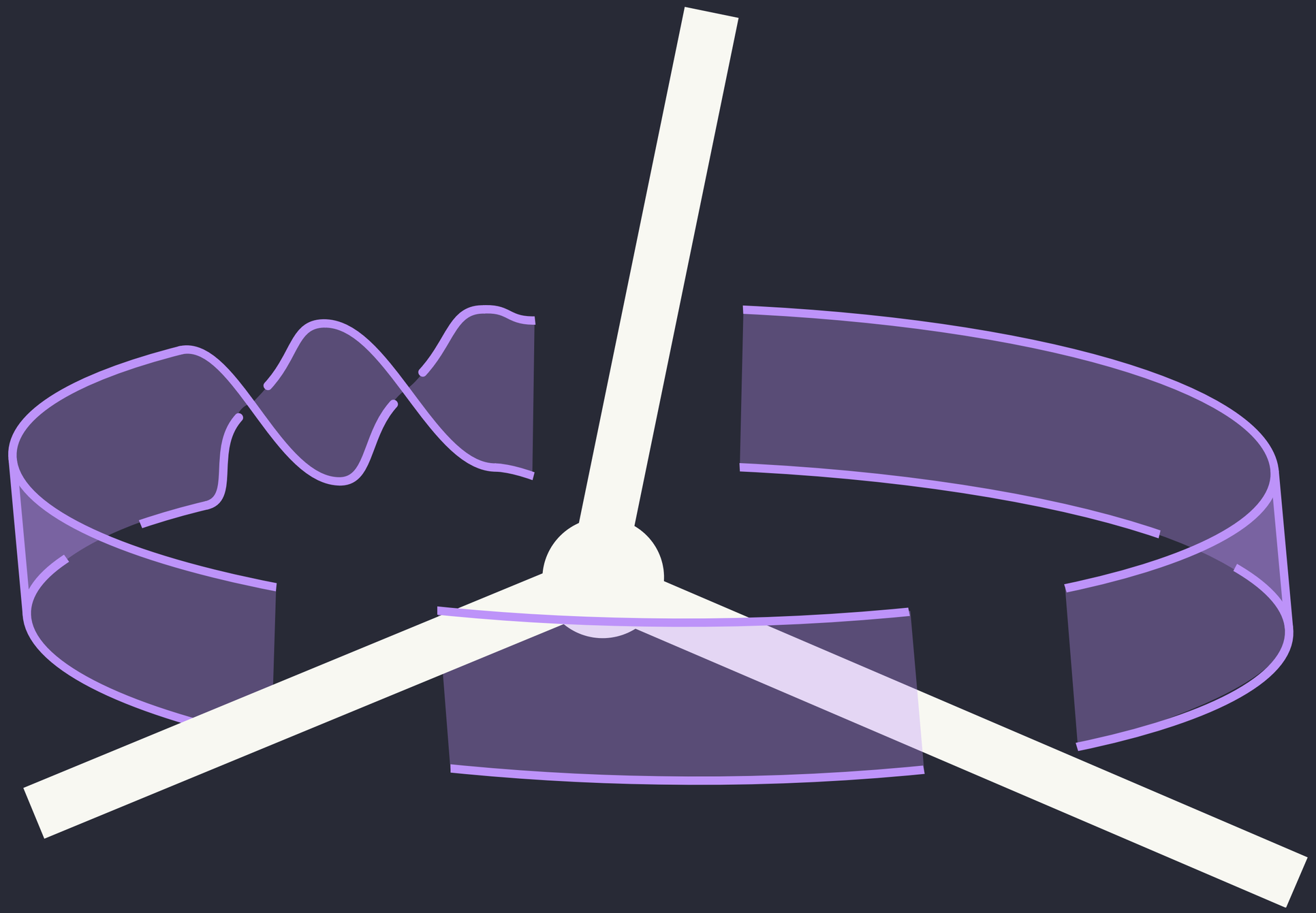
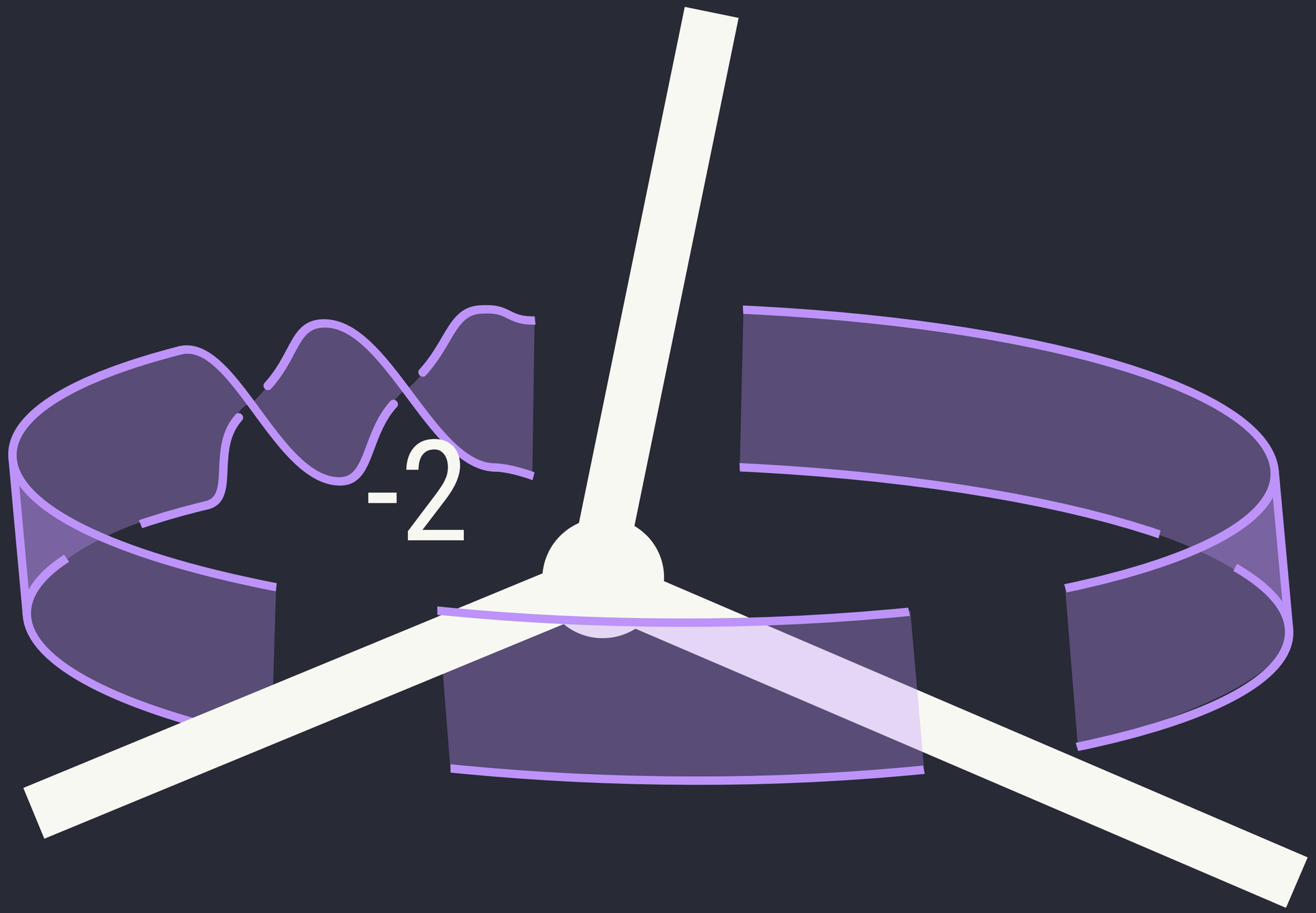
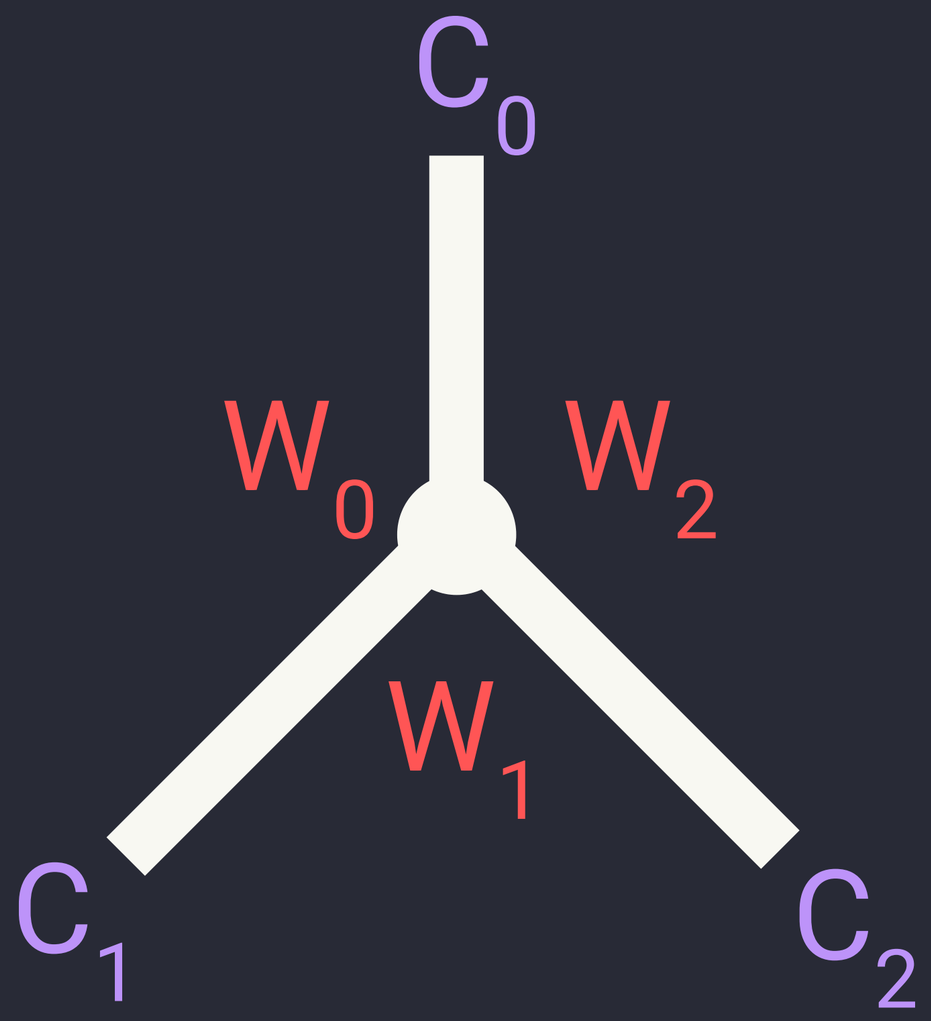
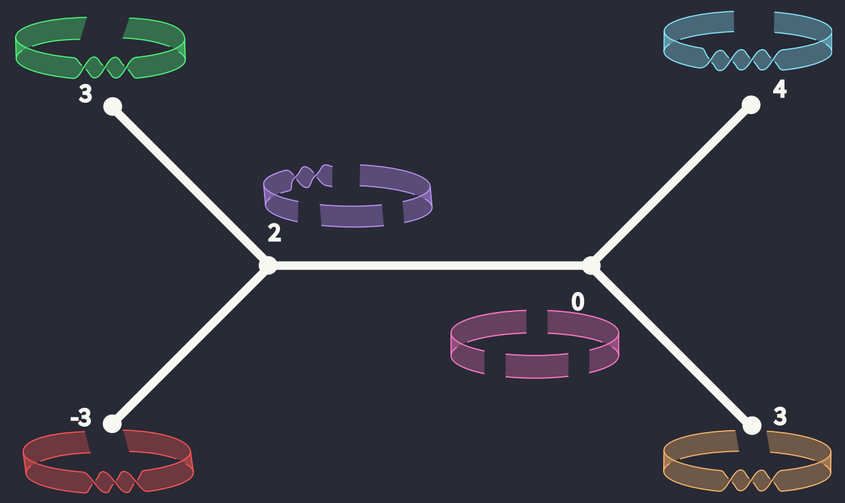



-2


C_0
W_0
W_2
W_1
C_1
C_2


3
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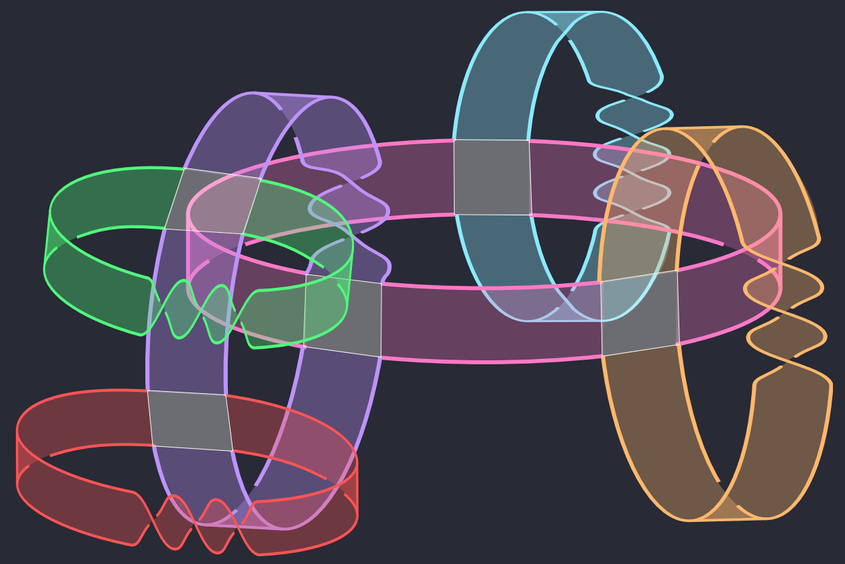
-3
3



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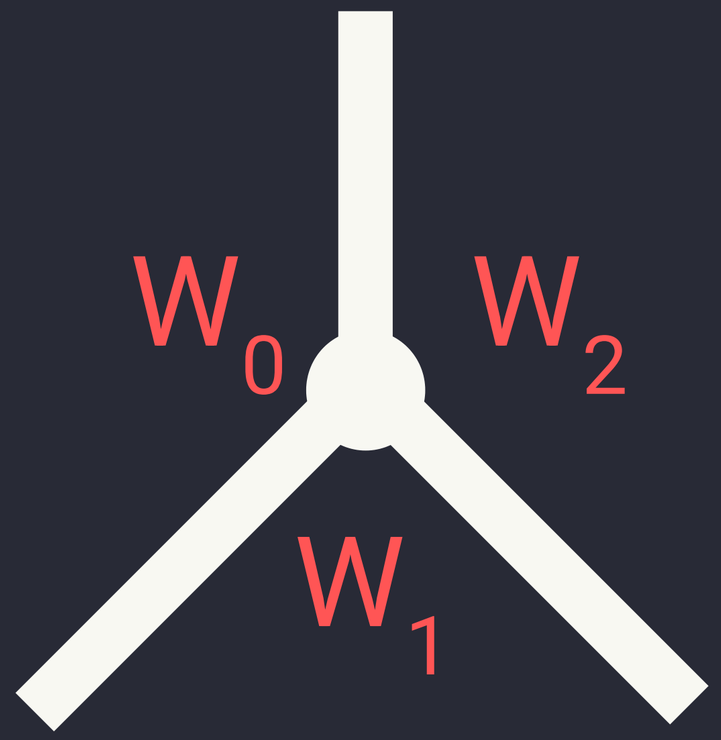
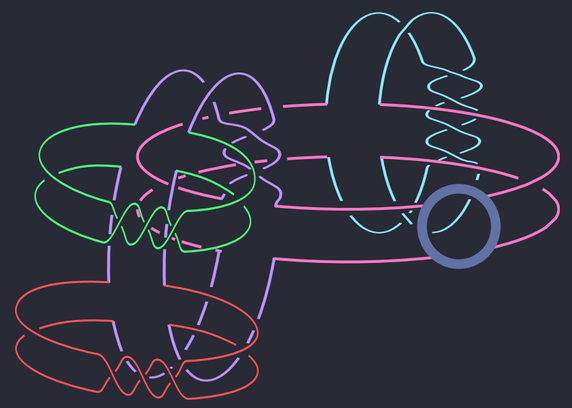
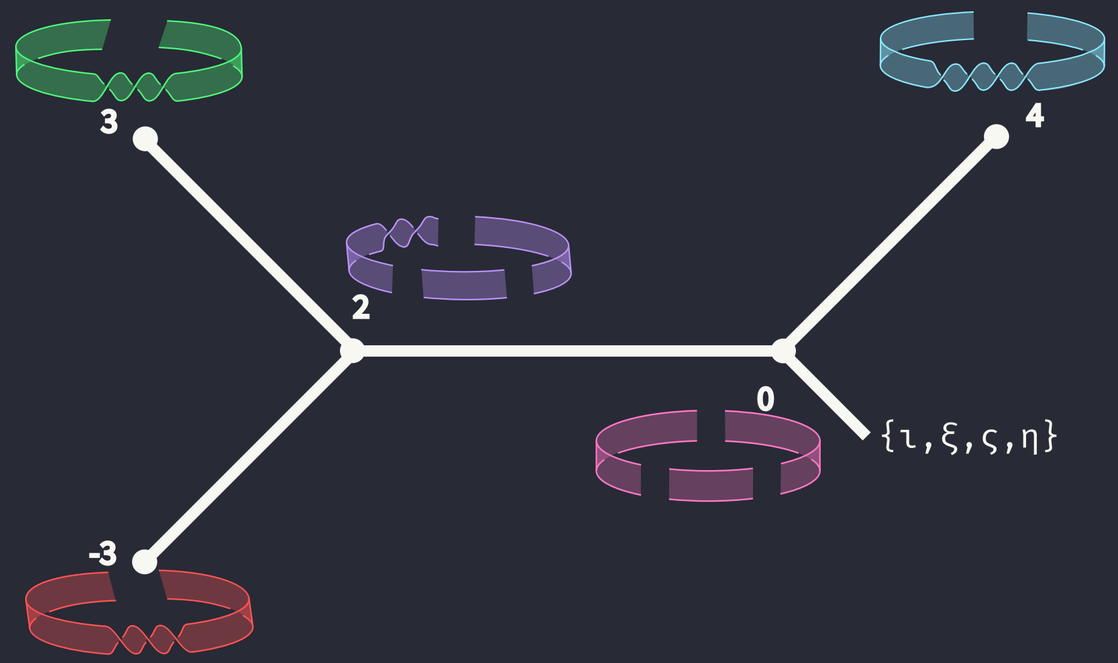
LOCAL VIEW OF A VERTEX

BONDS



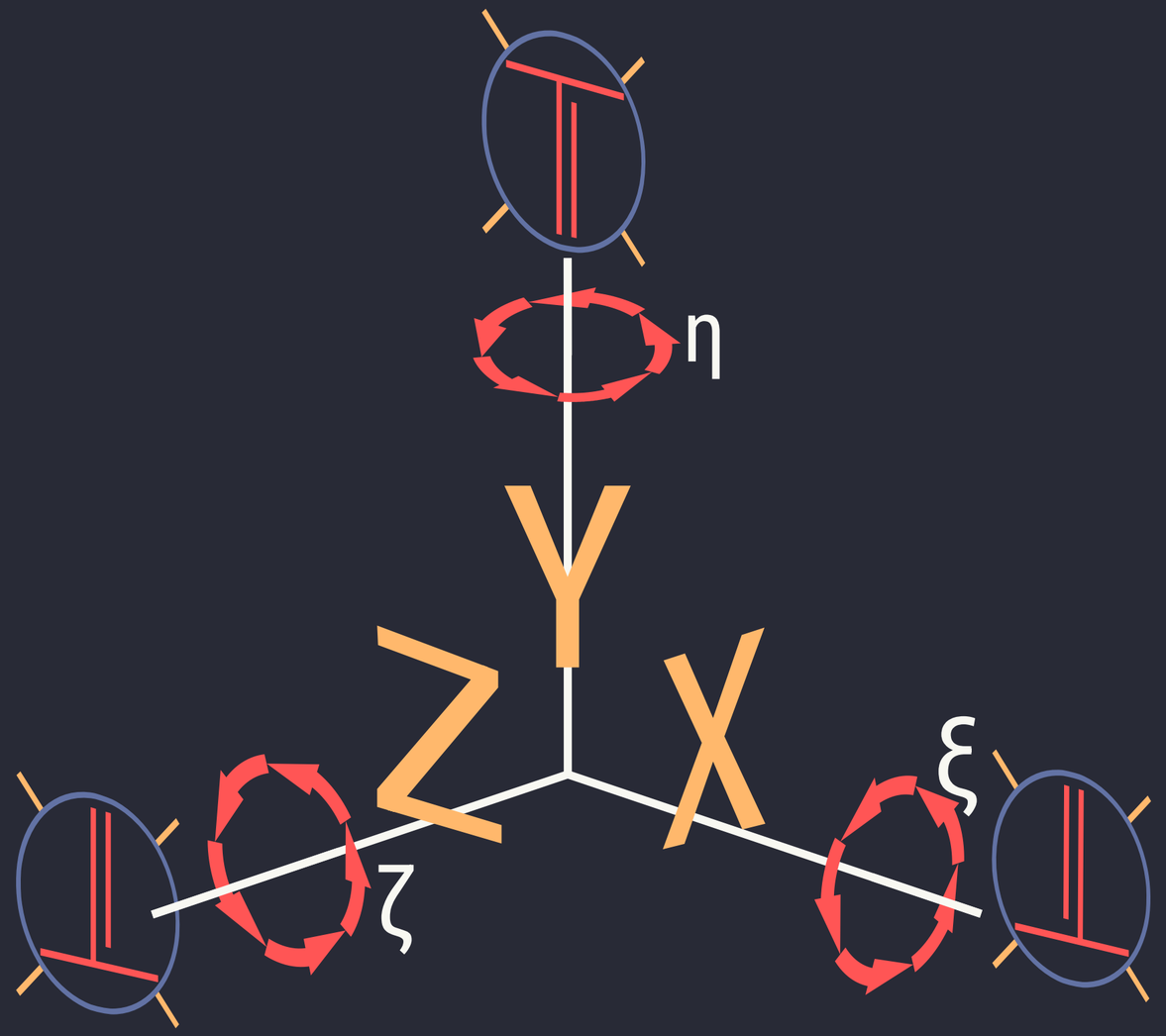
FREE BONDS





ROTATIONS

ELEMENTS OF KLEIN FOUR-GROUP (V_4)



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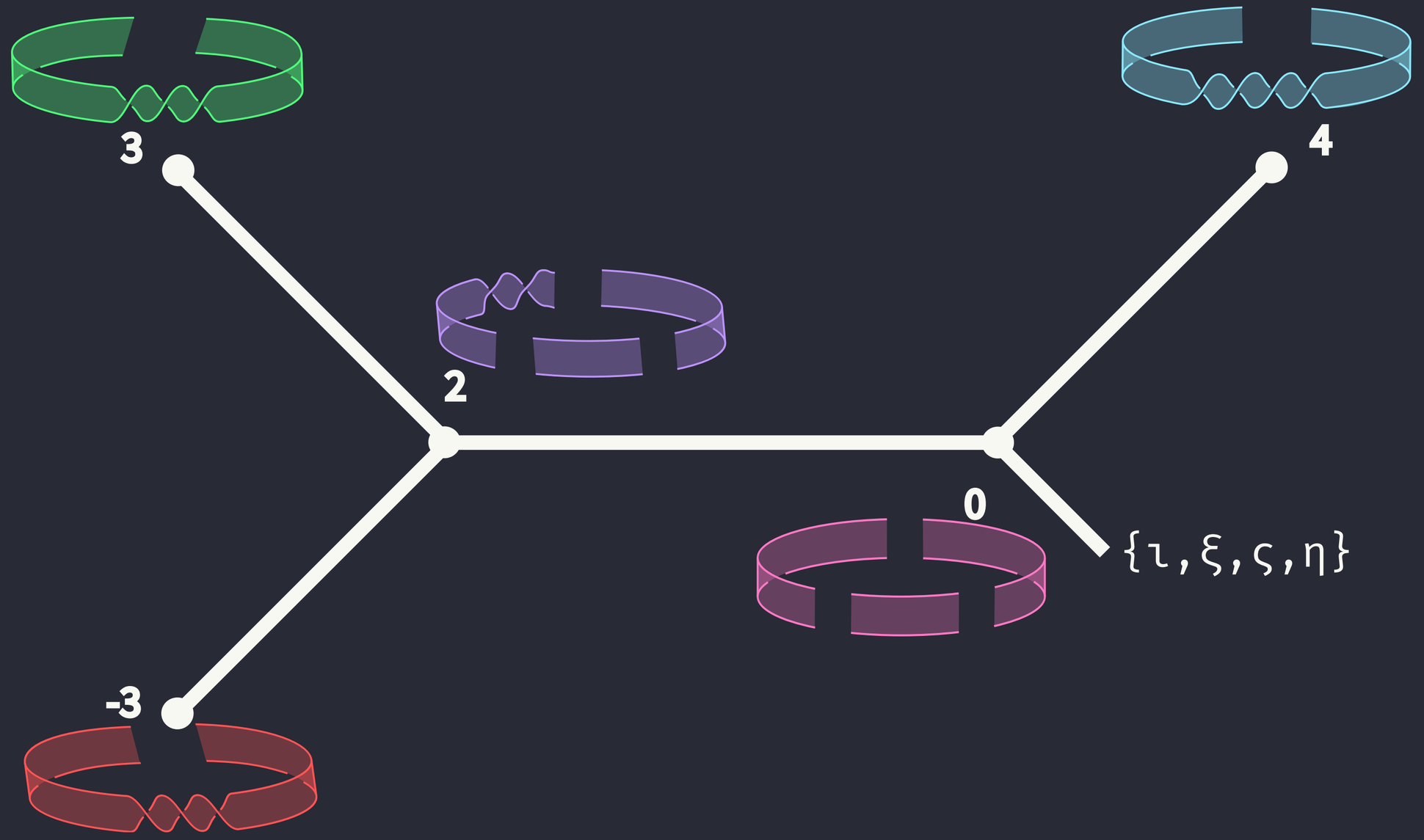
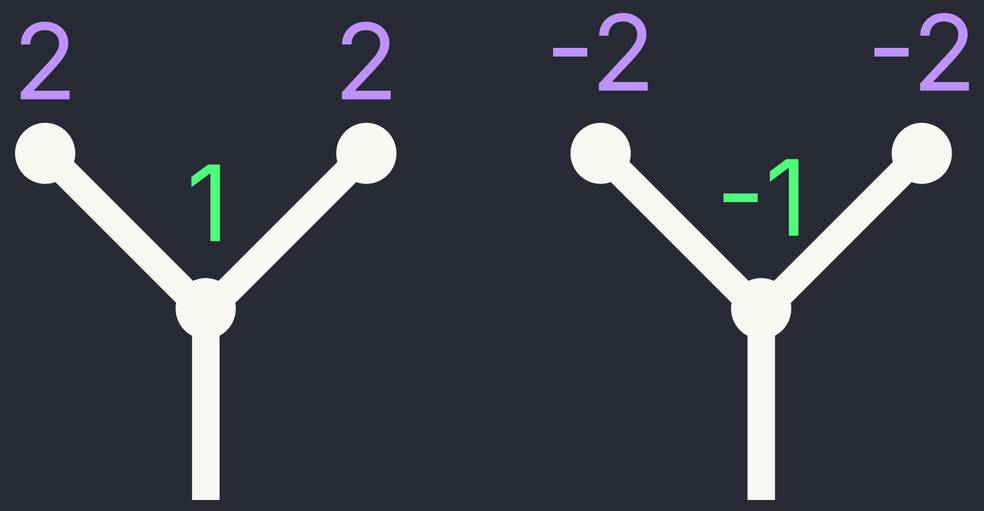
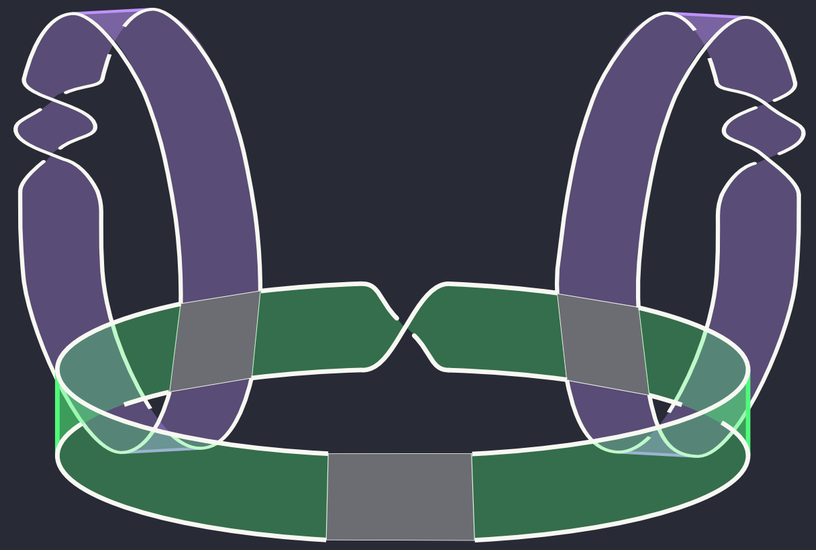
WEIGHTED PLANAR TANGLE TREE

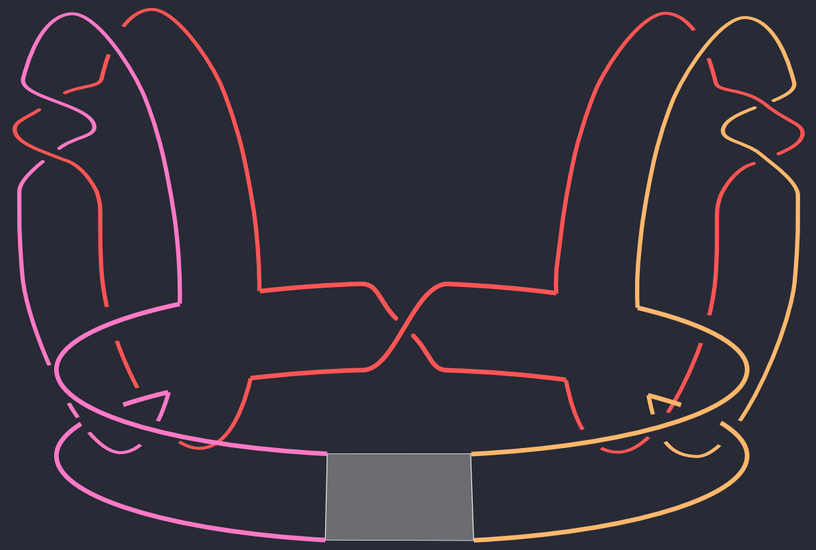


ANATOMY OF A WEIGHTED PLANAR TANGLE TREE

RINGS







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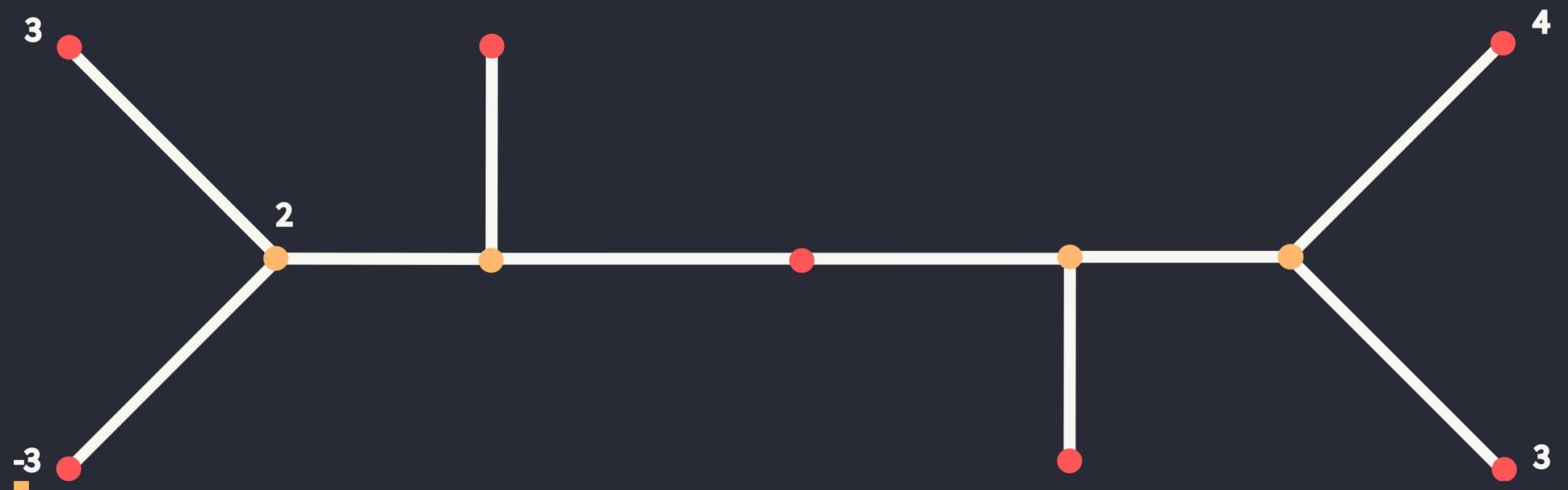

Definition

A vertex with valence ≥ 3 is called an *Essential Vertex*.

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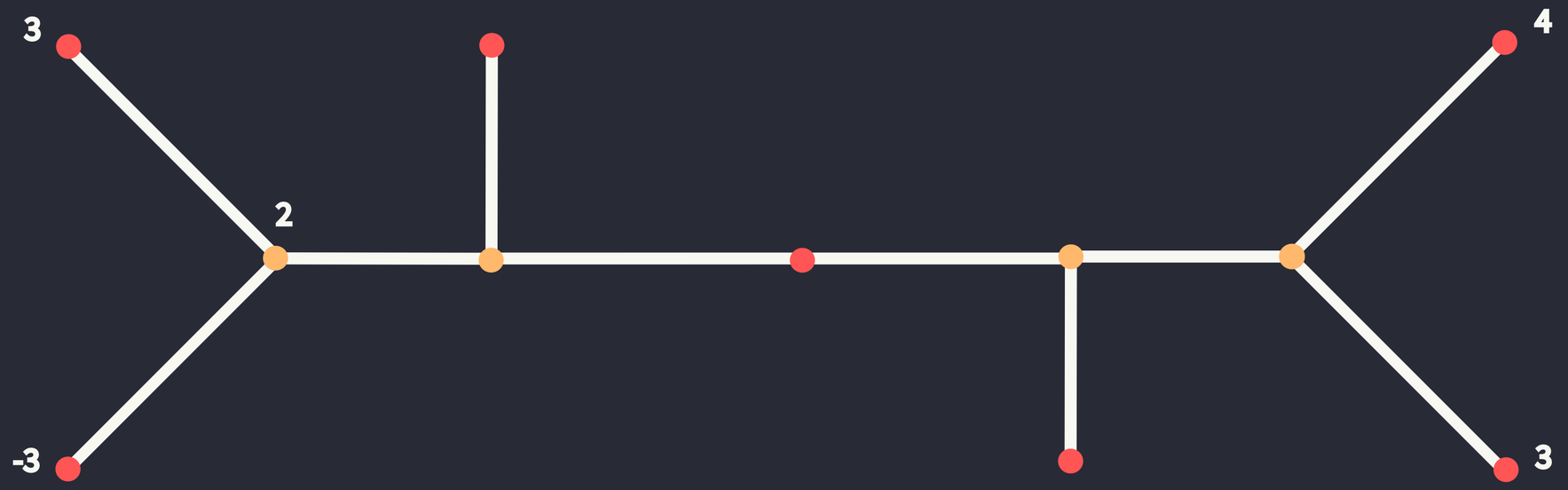
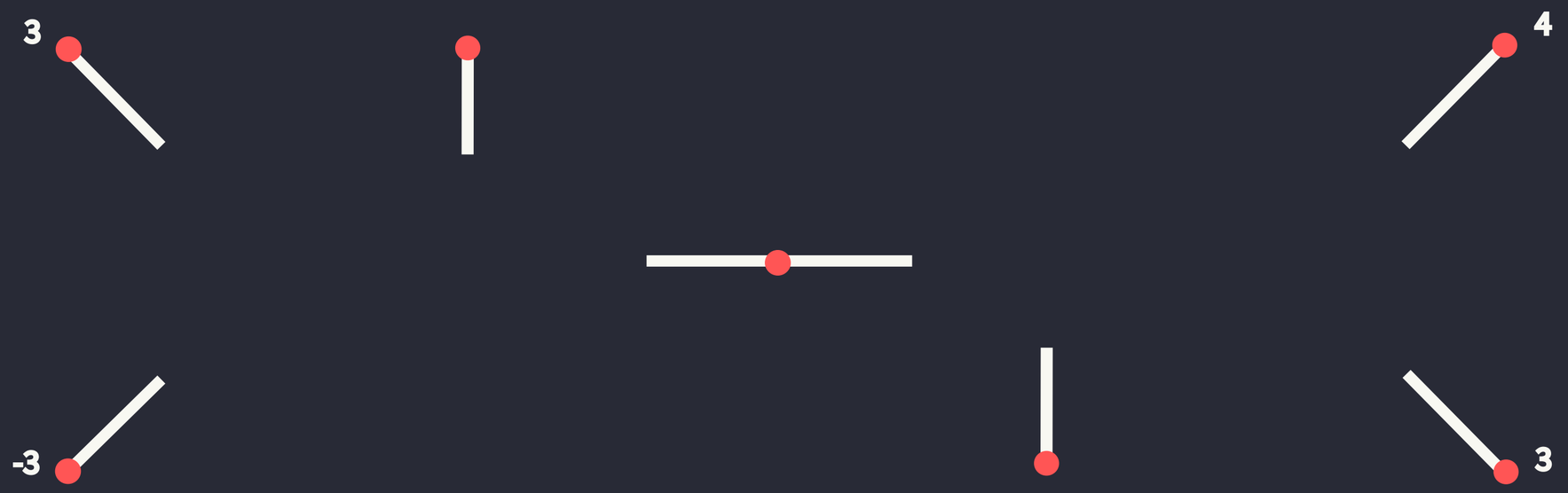
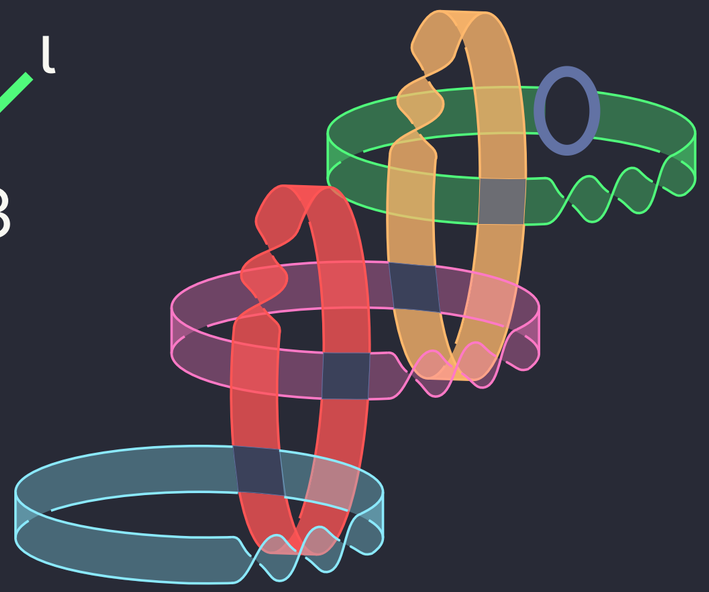


Definition

The subtrees remaining after excising all essential vertices and their bonds (half edges) are called the *Sticks* of a tree.



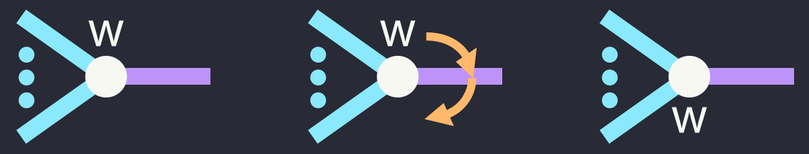
3
-2
3
-2
3



MOVES ON A WEIGHTED PLANAR TANGLE TREE

Definition: Part 1

The F_3' **move** on a weighted arborescent tree moves a weight w as in the image below and, if w is odd, reverse the cyclic order of weights and bonds at all vertices of the purple subtree lying at odd distance (count of edges between two vertices) from the vertex shown.



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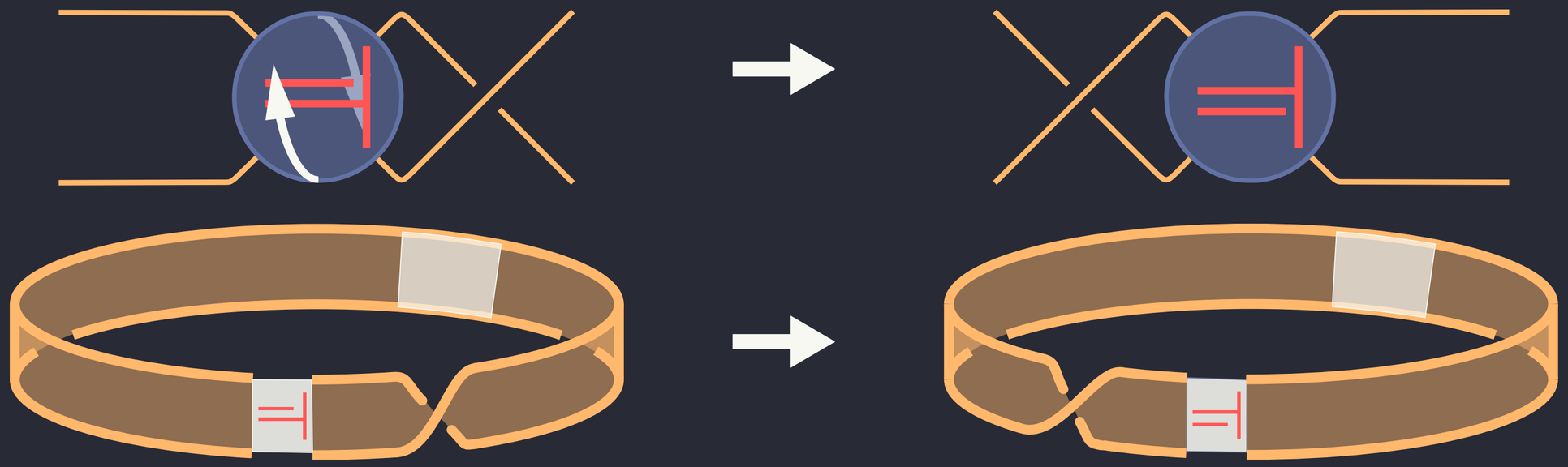
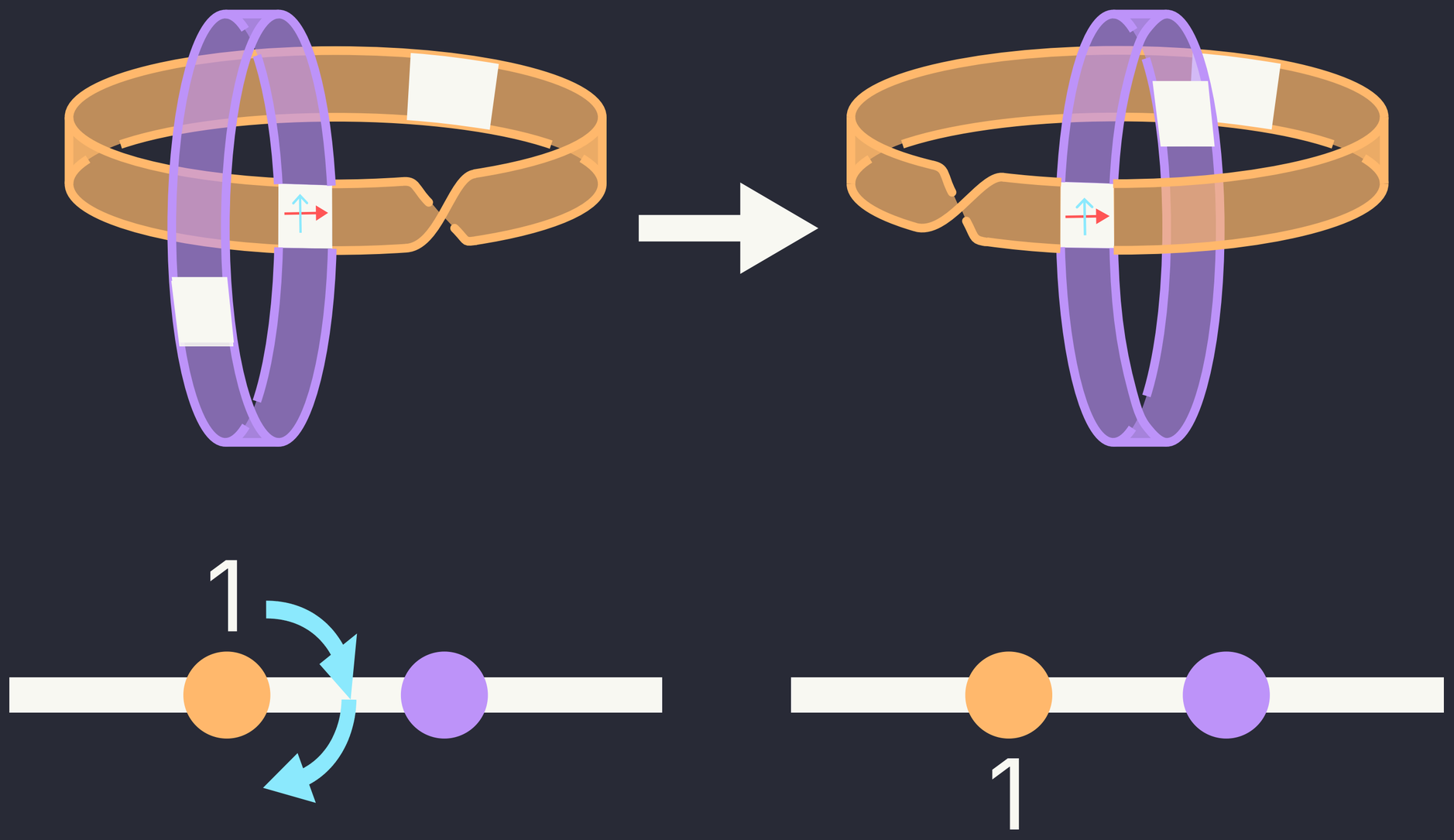
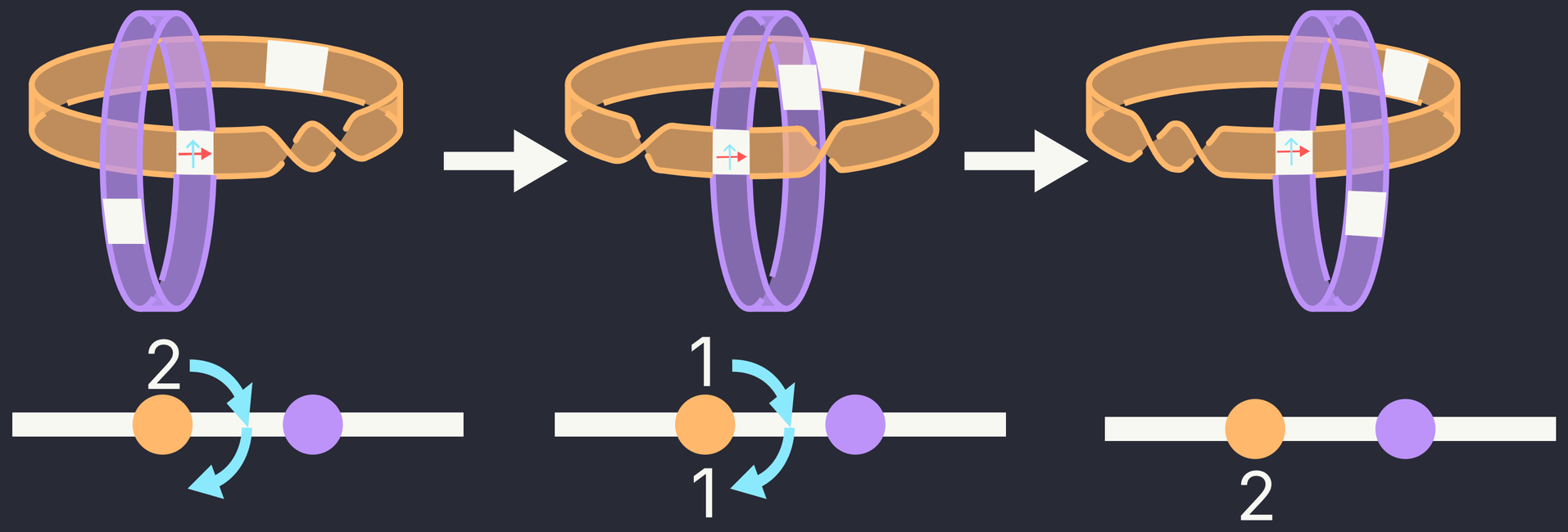
THE FLYPE



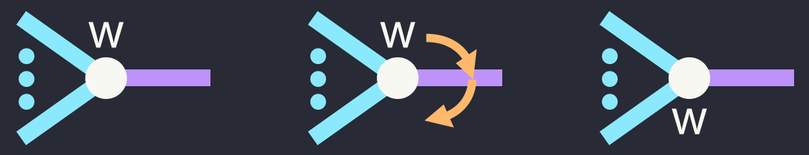
“if w is odd, reverse the cyclic order of weights and bonds at all vertices of the purple subtree lying at odd distance (count of edges between two vertices) from the vertex shown.”




2
1
1
2

Definition: Part 2

Also, when w is odd, apply ξ (X-axis rotation) to all free bonds in the purple subtree that are attached to a vertex at even distance from the vertex shown, and η (Y-axis rotation) to those at odd distance. The rotations are relative to the local orientations of the plumbing squares on the bands corresponding to vertices at even/odd distances from the vertex with weight w.



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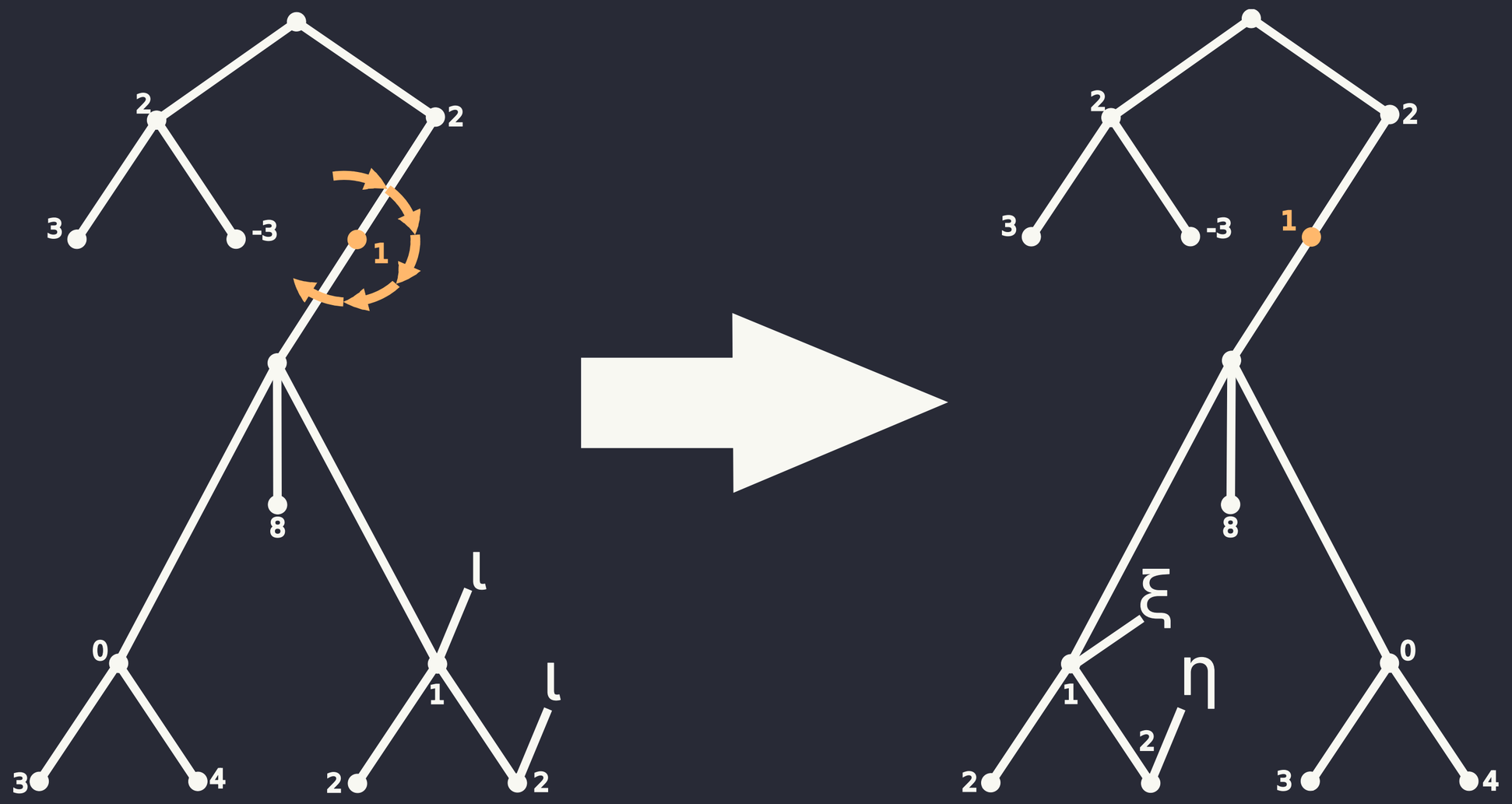
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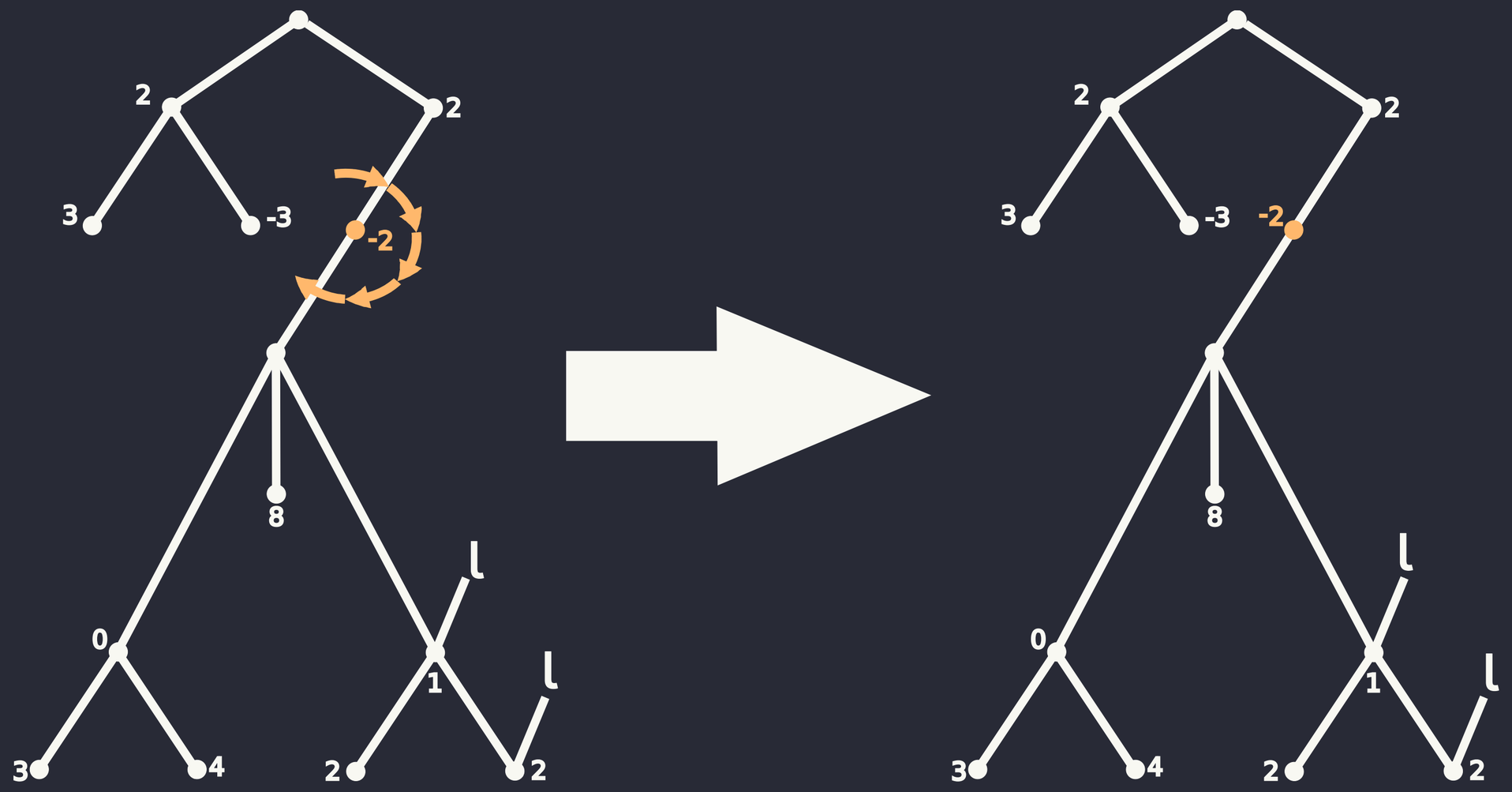
3
4


2
2
3
-3
-2
8
0
1
3
4
2
2
2
3
-3
-2
8
0
1
3
4
2
2

Definition

The F_2 **move** on a weighted arborescent tangle tree reverses the cyclic order of bonds and weights at one vertex on the tree and at every vertex at even distance from it; also apply η (Y-axis rotation) to every free bond of a vertex at even (or zero) distance, and apply ξ (X-axis rotation) to every free bond at odd distance.

F. Bonahon and L. Siebenmann, New geometric splittings of classical knots, and the classification and symmetries of arborescent knots, http://www-bcf.usc.edu/~fbonahon/Research/Publications.html

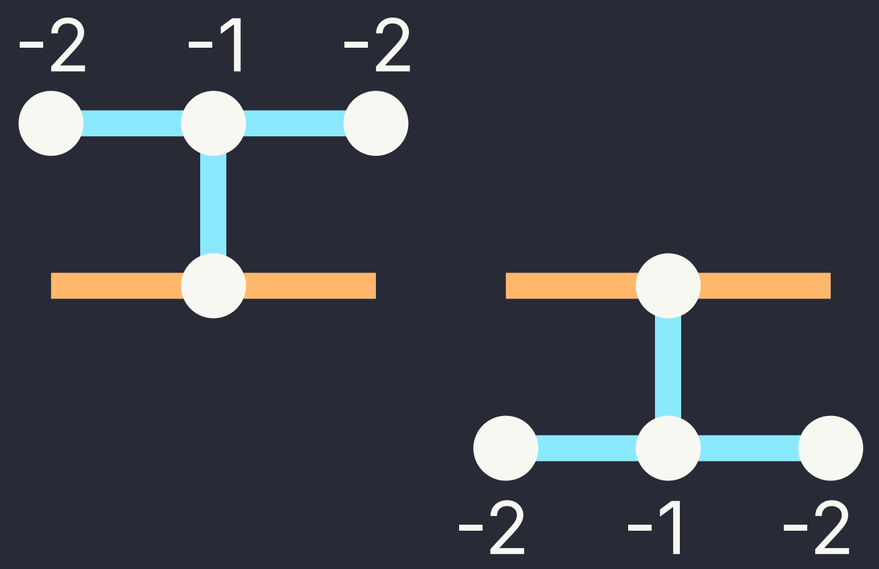
Definition

The F_1 **move** on a weighted arborescent tangle tree reverses the cyclic order of bonds and weights at every vertex of the graph and applies ζ (Z-axis rotation) to every free bond.

F. Bonahon and L. Siebenmann, New geometric splittings of classical knots, and the classification and symmetries of arborescent knots, http://www-bcf.usc.edu/~fbonahon/Research/Publications.html

Definition

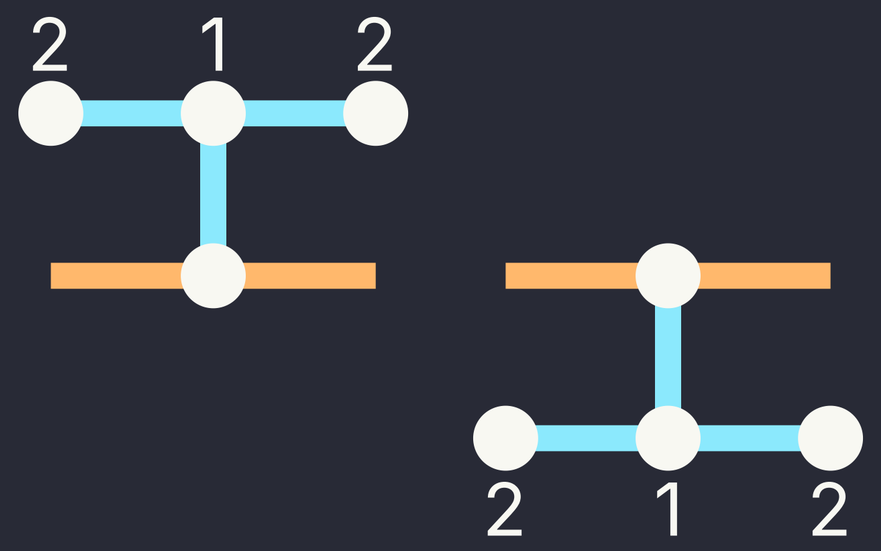
The R^- replaces the left image below with the right, leaving the rest of the tree unchanged.



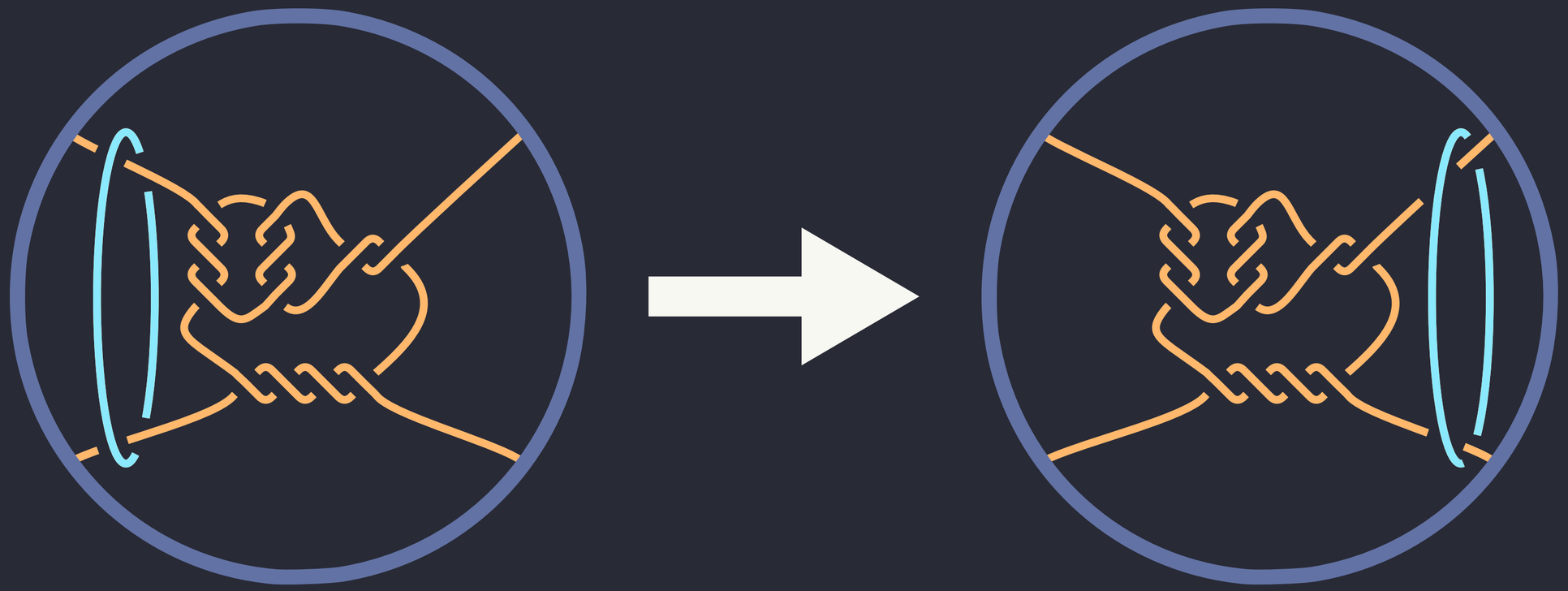
F. Bonahon and L. Siebenmann, New geometric splittings of classical knots, and the classification and symmetries of arborescent knots, http://www-bcf.usc.edu/~fbonahon/Research/Publications.html

Definition

The R^+ replaces the left image below with the right, leaving the rest of the tree unchanged.



F. Bonahon and L. Siebenmann, New geometric splittings of classical knots, and the classification and symmetries of arborescent knots, http://www-bcf.usc.edu/~fbonahon/Research/Publications.html



CANONICAL TREES

CANONICAL TREES

For our purposes, a weighted planar tree Γ is called a **canonical weighted planar tangle tree (CWPTT)** if it has a single free bond with a label from V_4 and satisfies the following conditions.

F. Bonahon and L. Siebenmann, New geometric splittings of classical knots, and the classification and symmetries of arborescent knots, http://www-bcf.usc.edu/~fbonahon/Research/Publications.html

WEIGHT CONDITION

At each vertex of Γ, at most one weight is non-zero.

F. Bonahon and L. Siebenmann, New geometric splittings of classical knots, and the classification and symmetries of arborescent knots, http://www-bcf.usc.edu/~fbonahon/Research/Publications.html

STICK CONDITIONS

1. On any stick the weights of the vertices are non-zero except for end vertices that have a bond free in Γ and for the case Γ is:

or

2. The non-zero weights along any stick are of alternating sign.
3. No end vertex of a stick has weight ± 1 unless it has a bond free in Γ.

F. Bonahon and L. Siebenmann, New geometric splittings of classical knots, and the classification and symmetries of arborescent knots, http://www-bcf.usc.edu/~fbonahon/Research/Publications.html

ONE OF

POSITIVITY CONDITION $(+)$-CWPTT

Except for those with a free bond, there are no sticks in Γ of the forms

or

OR

NEGATIVITY CONDITION $(-)$-CWPTT

Except for those with a free bond, there are no sticks in Γ of the forms

or

Theorem: Bonahon and Seibenmann

There exists an effective algorithm which, for any weighted planar tree Γ with free bonds labeled by elements of V_4, alters Γ by a sequence of moves of the calculus of arborescent tangles to produce a collection of positively (or negatively) canonical weighted planar trees.

F. Bonahon and L. Siebenmann, New geometric splittings of classical knots, and the classification and symmetries of arborescent knots, http://www-bcf.usc.edu/~fbonahon/Research/Publications.html

Theorem: Bonahon and Seibenmann

Consider two positive (or negative) CWPTT Γ and Γ', with free bonds labeled by elements of V_4. Plumbing, according to Γ and Γ' gives isomorphic arborescent tangles if and only if Γ and Γ' can be deduced from each other by a sequence of moves (F_1), (F_2), (F_3'), and the modified ring moves $(\pm R)$.

F. Bonahon and L. Siebenmann, New geometric splittings of classical knots, and the classification and symmetries of arborescent knots, http://www-bcf.usc.edu/~fbonahon/Research/Publications.html

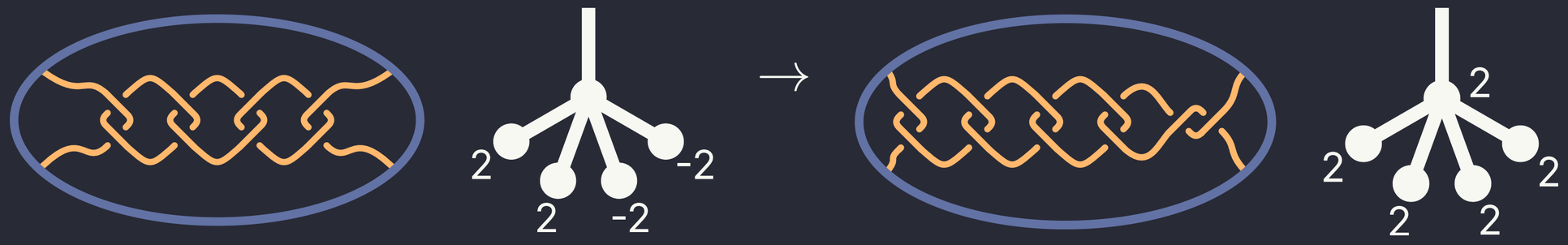
Definition

For weighted planar tangle tree Γ, with weights $\{w_i\}$. We call

$$\mathrm{TCN} = \sum |w_i|$$

the **Tree Crossing Number (TCN)**.

CWPTT ARE NOT MINIMAL



A CANONICAL VERTEX [S]

A vertex v_i of a weighted planar tangle tree Γ with a single free bond labeled from V_4 is said to be $(+)$**-canonical** if v_i has at most one non-zero weight w_i and i is:

- Zero (v_0 the root) with no further conditions.
- Non-zero with the following conditions satisfied:

I. If the valence of v_i is 1 then all of:

- (Stick Condition)
    1. $w_i \neq 0$ unless $i = 1$ and $w_0 = 0$ (the weight of the root)
    2. $w_i \neq \pm 1$
    3. If the valence of v_{i-1} (the parent) is 2 then $\operatorname{sign}(w_i) \neq \operatorname{sign}(w_{i-1})$ unless $i = 1$ and $w_0 = 0$
- (Positivity Condition)
    1. If the valence of v_{i-1} (the parent) is greater than 2 then $w_i \neq -2$

Stick Condition

1. On any stick the weights of the vertices are non-zero except for end vertices that have a bond free in Γ and for the case Γ is:

2. The non-zero weights along any stick are of alternating sign.
3. No end vertex of a stick has weight ± 1 unless it has a bond free in Γ.

I. If the valence of v_i is 1 then all of:

- (Stick Condition)
    1. $w_i \neq 0$ unless $i = 1$ and $w_0 = 0$ (the weight of the root)
    2. $w_i \neq \pm 1$
    3. If the valence of v_{i-1} (the parent) is 2 then $\operatorname{sign}(w_i) \neq \operatorname{sign}(w_{i-1})$ unless $i = 1$ and $w_0 = 0$

Positivity Condition:

Except for those with a free bond, there are no sticks in Γ of the forms

or

I. If the valence of v_i is 1 then all of:

- (Positivity Condition)
    1. If the valence of v_{i-1} (the parent) is greater than 2 then $w_i \neq -2$

II. If the valence of v_i is 2 then all of:

- (Stick Condition)
    1. $w_i \neq 0$
    2. If valence of the parent or valence of the child is greater than 2 (is essential) then $w_i \neq \pm 1$
    3.
        1. If valence of the parent is 2 then $\operatorname{sign}(w_i) \neq \operatorname{sign}(w_{i-1})$ (the parent)
        2. If valence of the child is 2 then $\operatorname{sign}(w_i) \neq \operatorname{sign}(w_{i+1})$ (the child)
- (Positivity Condition)
    1. If valence of the parent and valence of the child is greater than 2 (both are essential) then $w_i \neq -2$

Stick Condition

1. On any stick the weights of the vertices are non-zero except for end vertices that have a bond free in Γ and for the case Γ is:

or

2. The non-zero weights along any stick are of alternating sign. 3. No end vertex of a stick has weight ± 1 unless it has a bond free in Γ.

II. If the valence of v_i is 2 then all of:

- (Stick Condition)
    1. $w_i \neq 0$
    2. If valence of the parent or valence of the child is greater than 2 (is essential) then $w_i \neq \pm 1$
    3.
        1. If valence of the parent is 2 then $\operatorname{sign}(w_i) \neq \operatorname{sign}(w_{i-1})$ (the parent)
        2. If valence of the child is 2 then $\operatorname{sign}(w_i) \neq \operatorname{sign}(w_{i+1})$ (the child)

Positivity Condition: Except for those with a free bond, there are no sticks in Γ of the forms

or

II. If the valence of v_i is 2 then all of:

- (Positivity Condition)
    1. If valence of the parent and valence of the child is greater than 2 (both are essential) then $w_i \neq -2$

Theorem [S]

Γ is a $(+)$-CWPTT if and only if all the vertices of Γ are $(+)$-canonical.

TABULATION OF ARBORESCENT TANGLES

WHAT DO WE HAVE?

1. A combinatorial structure

1. A combinatorial structure
2. A way to distinguish two arborescent tangles

WHAT WE DON’T HAVE?

1. A unique representative

1. A unique representative
2. An efficient storage strategy

1. A unique representative
2. An efficient storage strategy
3. An efficient generation strategy

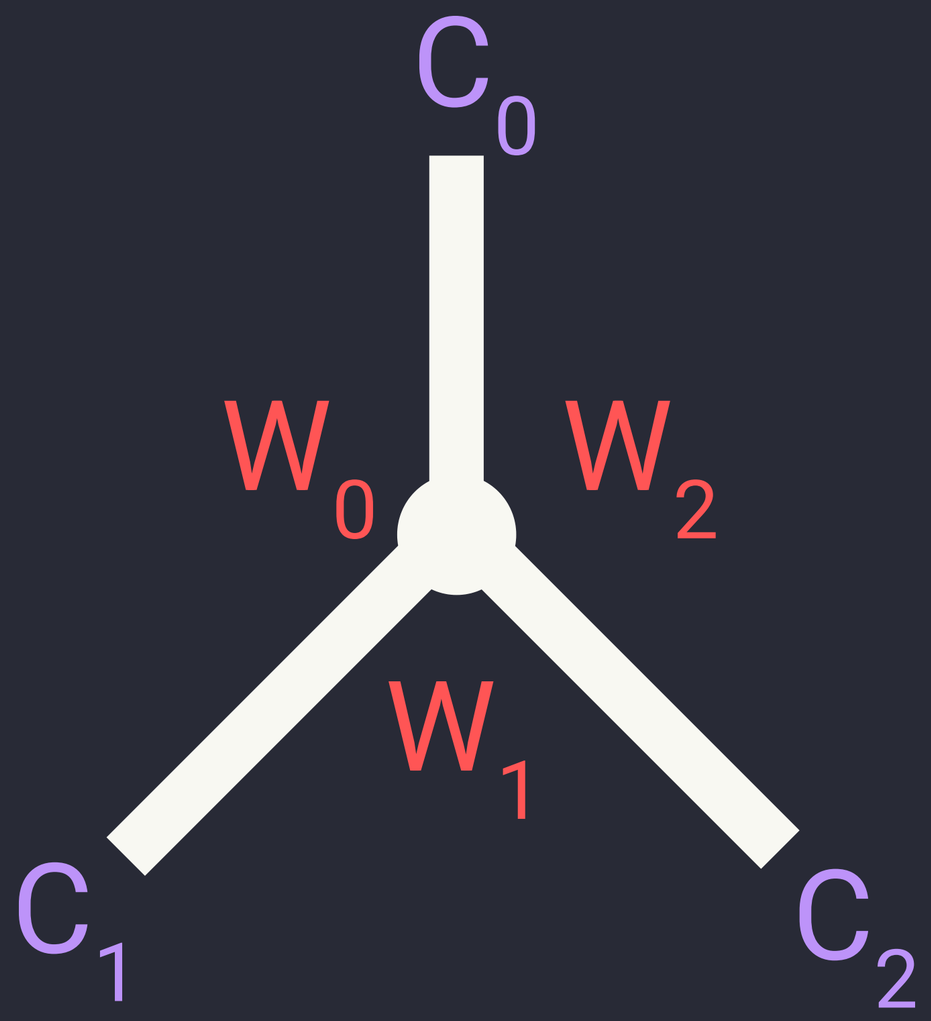
A UNIQUE REPRESENTATIVE

Definition [S]

A CWPTT is called **right leaning** if all weights are in the highest indexed region of each vertex. Additionally, any ring subtrees that are children of a vertex are the highest indexed children of that vertex.

Note

Not using “abbreviated trees” makes generation a bit less complex.


C_0
W_0
W_2
W_1
C_1
C_2

Theorem [S]

Every arborescent tangle has a right leaning CWPTT representative.

Definition [S]

A CWPTT is called an **identity tree** if its free bond is marked by $\iota \in V_4$.

Theorem [S]

Every arborescent tangle has an identity CWPTT representative.

Definition [S]

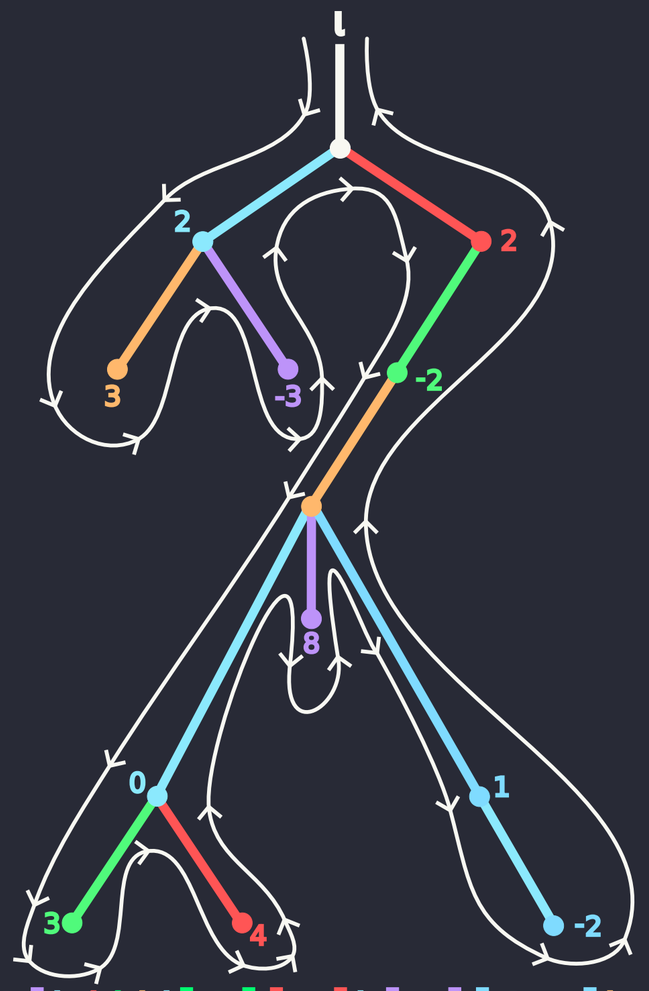
A CWPTT is called a **right leaning identity tangle tree (RLITT)** if it's a right leaning and identity tree.

Theorem [S]

Every $(+)$-CWPTT has a unique right leaning identity representative.

1. ✓ A unique representative
2. An efficient storage strategy
3. An efficient generation strategy

AN EFFICIENT STORAGE STRATEGY



ι((2[3][-3])((((([3][4])[8][2 1])-2)2))

Cayley, A. (1857). ON THE THEORY OF THE ANALYTICAL FORMS CALLED TREES (The collected mathematical papers Vol. 3). Cambridge University Press.
https://doi.org/10.1017/CBO9780511703690

1. ✓ A unique representative
2. ✓ An efficient storage strategy
3. An efficient generation strategy

AN EFFICIENT GENERATION STRATEGY

Definition [S]

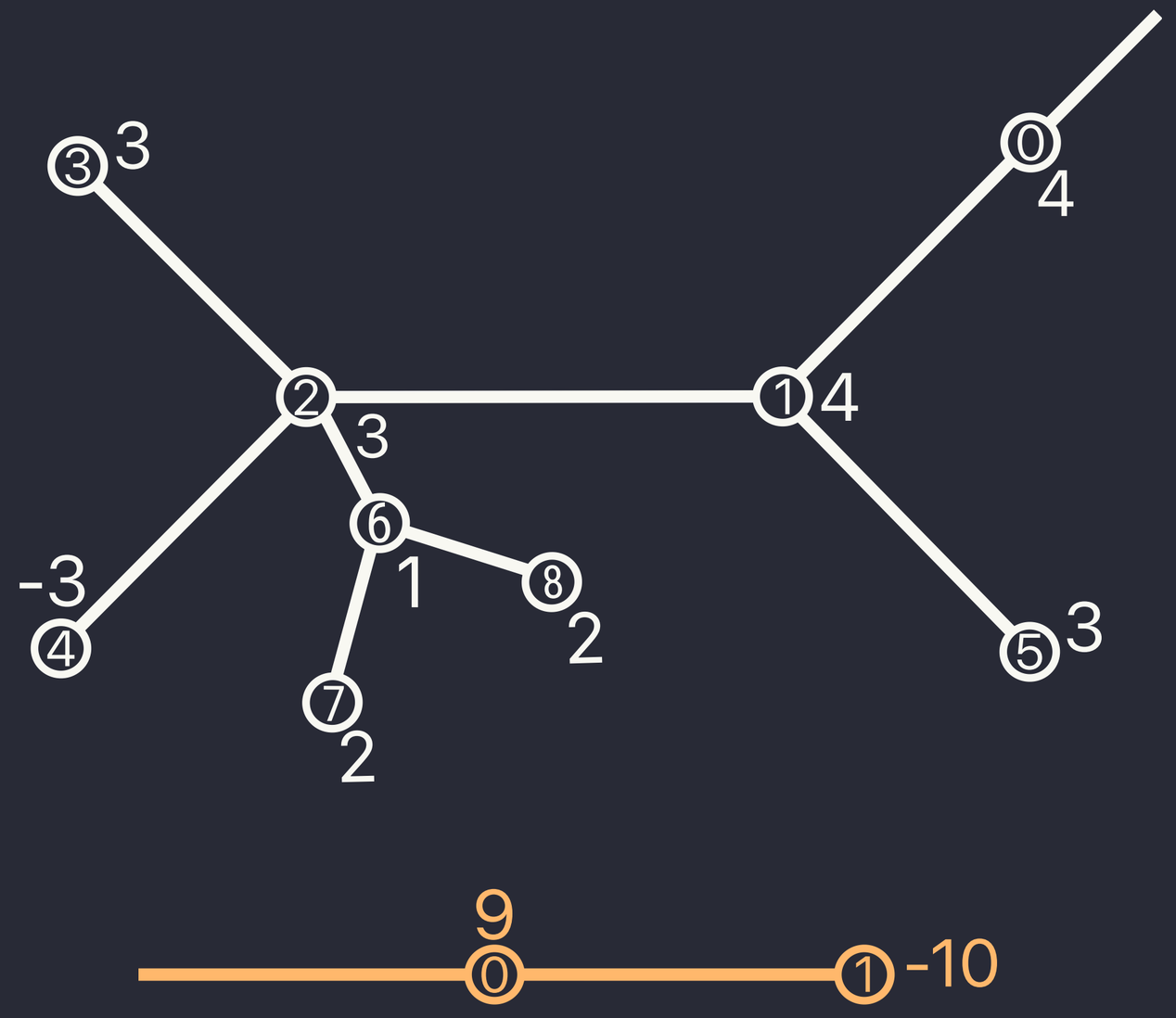
For WPTT Γ_r and Γ_s, define the **grafting operation**

$$\Gamma_r \times \Gamma_s \mapsto \Gamma_r \star_i \Gamma_s$$

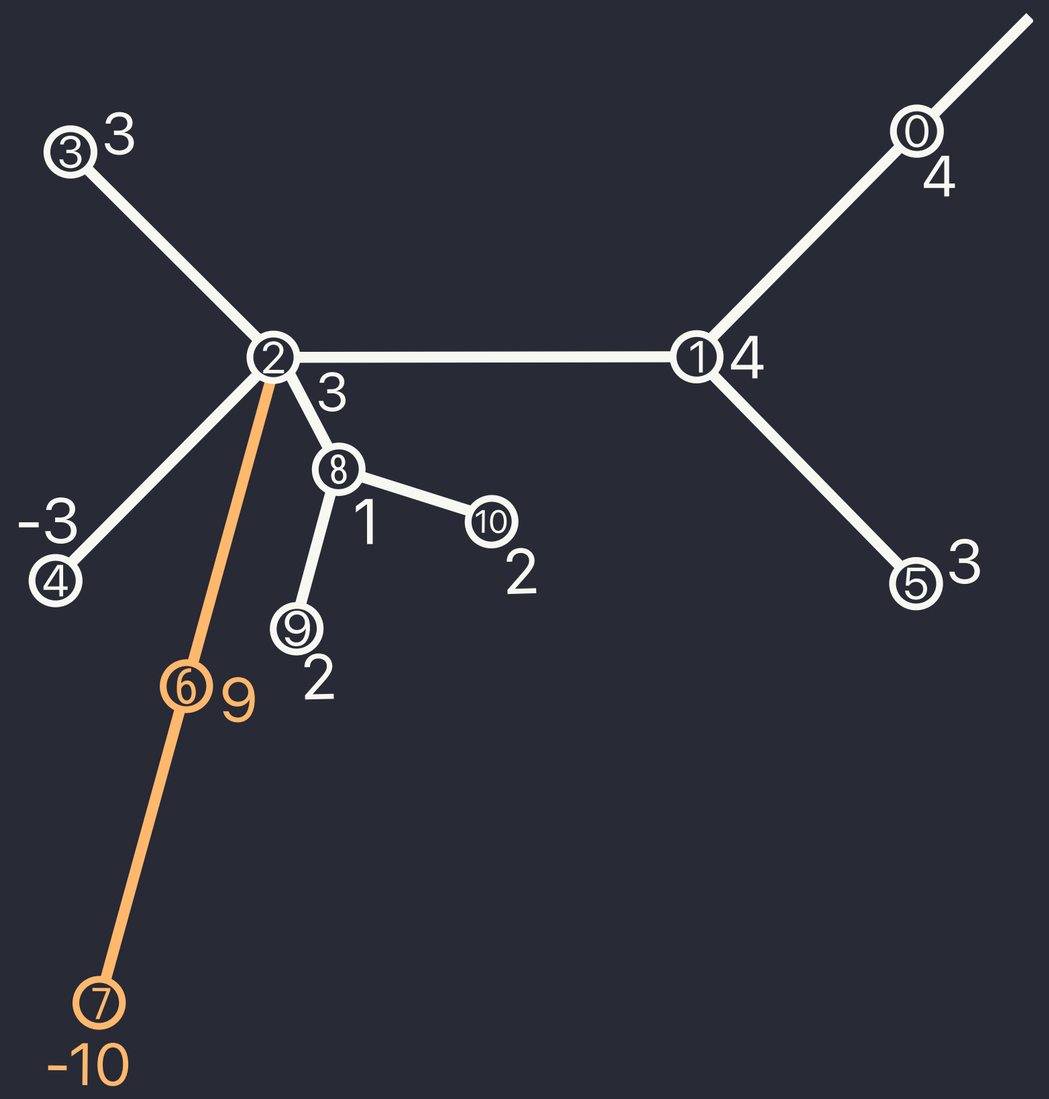
as follows. At the vertex v_i of a Γ_r, introduce a bond connecting to the free bond at the root of Γ_s, reindexing as needed. We also require that Γ_s be grafted so that the rightmost weight of v_i and any ring subtrees of v_i remain to the right of the scion after grafting.

When grafting at the root v_0 we omit the index label in the grafting operation, that is, $\star_0$ is written simply as $\star$. We call Γ_r the **rootstock** and Γ_s the **scion**.

Γ_r and Γ_s



$\Gamma_r \star_2 \Gamma_s$



Theorem [S]

Every $\Gamma\ (+)$-RLITT of TCN n is one of two forms:

1. Γ is a single vertex with weight $\pm n$.
2. Γ is the result of grafting at the root of some rootstock Γ_r and $(+)$-RLITT scion Γ_s where:
    1. In Γ_r, v_0 is valence two, and v_1 is canonical except for violating the stick condition by $\mathrm{Sign}\ (v_0) = \mathrm{Sign}\ (v_1)$. Each vertex in $\{v_i\}_{i=2}^{n}$ of Γ_r is $(+)$-canonical.
    2. Γ_r is $(+)$-RLITT

1
-2
3
3

-1
-2
3
3

Algorithm [S]

Input

- A collection of of RLITT scions T_s
- A collection of RLITT rootstocks T_r

Output

- A collection of weighted planar trees

Routine

1. for each combination of $\Gamma_r \in T_r$ and $\Gamma_s \in T_s$
    1. Compute $\Gamma = \Gamma_r \star \Gamma_s$
    2. Report Γ
    3. Continue to the iteration of the loop if v_0 in Γ_r is valence other than two
        1. Set v_0 to $-v_0$ for Γ
        2. Report Γ

GUARANTEE RLITT

1. Weight Condition
2. Identity Condition
3. Right Leaning Condition
4. Stick Condition
5. Positivity/Negativity Condition

GUARANTEE RLITT

1. ✓ Weight Condition
2. ✓ Identity Condition
3. ✓ Right Leaning Condition
4. Stick Condition
5. Positivity/Negativity Condition

STICK CONDITION

Theorem [S]

For Γ_r a $(+)-$RLITT and Γ_s a $(+)-$RLITT scion, the result of $\Gamma = \Gamma_r \star \Gamma_s$ is canonical, if all v_i at distance 1 or less from the root are canonical.

Definition [S]

A $(+)-$RLITT (respectively $(-)-$RLITT) Γ with root weight w_0 is called a **good scion** when either:

1. $w_0 \neq 0$
2. $w_0 = 0$ and the valence of v_0 is greater than 2 (essential)

Algorithm [S]

Input

- A collection of RLITT good scions T_s
- A collection of RLITT rootstocks T_r

Output

- A collection of weighted planar trees (still not guaranteed to be RLITT)

Routine

1. for each combination of $\Gamma_s \in T_s$ and $\Gamma_r \in T_r$
    1. Compute $\Gamma = \Gamma_r \star \Gamma_s$
    2. for each vertex v_i at distance 1 from the root of Γ
        1. Continue to the next iteration of the outer loop if v_i fails to satisfy the stick condition
    3. Report Γ
    4. Continue to the iteration of the loop if v_0 in Γ_r is valence other than two
        1. Set v_0 to $-v_0$ for Γ
        2. Report Γ

GUARANTEE RLITT

1. ✓ Weight Condition
2. ✓ Identity Condition
3. ✓ Right Leaning Condition
4. ✓ Stick Condition
5. Positivity/Negativity Condition

POSITIVITY/NEGATIVITY CONDITION

Theorem [S]

For Γ_r a non-negative $(+)-$RLITT, and Γ_s a good non-positive $(-)-$RLITT scion, the result of $\Gamma = \Gamma_r \star \Gamma_s$ is non-canonical.

Algorithm: Generate $(+)$-RLITT [S]

Input

- A collection of $(+)$-RLITT good scions T_s
- A collection of $(+)$-RLITT rootstocks T_r

Output

- A collection of weighted planar trees (still not guaranteed to be RLITT)

Routine

1. for each combination of $\Gamma_s \in T_s$ and $\Gamma_r \in T_r$
    1. Compute $\Gamma = \Gamma_r \star \Gamma_s$
    2. for each vertex v_i at distance 1 from the root of Γ
        1. Continue to the next iteration of the outer loop if v_i fails to satisfy the stick condition
        2. Continue to the next iteration of the outer loop if v_i fails to satisfy the positivity condition
    3. Report Γ
    4. Continue to the iteration of the loop if v_0 in Γ_r is valence other than two
        1. Set v_0 to $-v_0$ for Γ
        2. Report Γ

Algorithm: Generate $(-)$-RLITT [S]

Input

- A collection of $(-)$-RLITT good scions T_s
- A collection of $(-)$-RLITT rootstocks T_r

Output

- A collection of weighted planar trees (still not guaranteed to be RLITT)

Routine

1. for each combination of $\Gamma_s \in T_s$ and $\Gamma_r \in T_r$
    1. Compute $\Gamma = \Gamma_r \star \Gamma_s$
    2. for each vertex v_i at distance 1 from the root of Γ
        1. Continue to the next iteration of the outer loop if v_i fails to satisfy the stick condition
        2. Continue to the next iteration of the outer loop if v_i fails to satisfy the positivity condition
    3. Report Γ
    4. Continue to the iteration of the loop if v_0 in Γ_r is valence other than two
        1. Set v_0 to $-v_0$ for Γ
        2. Report Γ

GUARANTEE RLITT

1. ✓ Weight Condition
2. ✓ Identity Condition
3. ✓ Right Leaning Condition
4. ✓ Stick Condition
5. ✓ Positivity/Negativity Condition

PARTITIONING JOBS

$$
\begin{aligned}
&(0, \mathrm{TCN}) \\
&(1, \mathrm{TCN} - 1) \\
&\vdots \\
&(\mathrm{TCN} - 1, 1) \\
&(\mathrm{TCN}, 0)
\end{aligned}
$$

Algorithm [S]

Input

- A target TCN n

Output

- A set T of all RLITT up to TCN

Routine

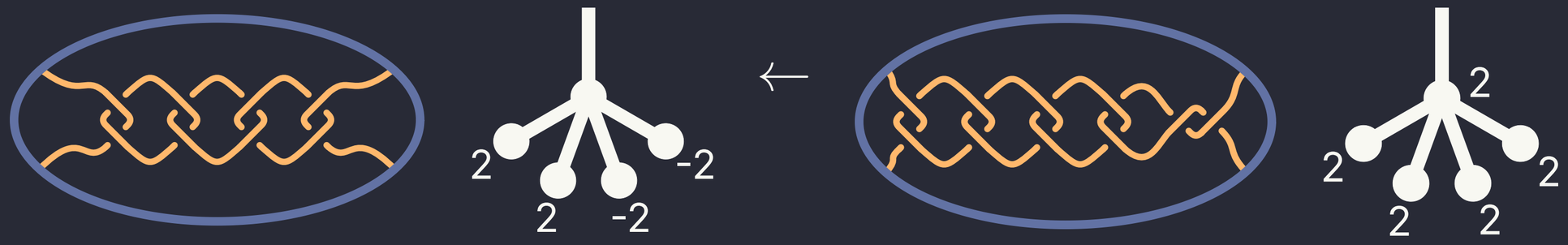
1. Set T to be the set $\{\iota[0],\ \iota[0\ 0],\ \iota[1],\ \iota[-1],\ \iota[2],\ \iota[-2],\ \cdots,\ \iota[n],\ \iota[-n],\ \}$
2. for i from 2 to TCN
    1. for j from i-2 to 1
        1. Set T_{r+} to be the set of $(+)$-RLITT with TCN equal to j
        2. Set T_{r-} to be the set of $(-)$-RLITT with TCN equal to j
        3. Set T_{s+} to be the set of $(+)$-RLITT good scions with TCN equal to $n-j$
        4. Set T_{s-} to be the set of $(-)$-RLITT good scions with TCN equal to $n-j$
        5. Execute “Generate $(+)$-RLITT” input T_{r+} and T_{s+}
        6. Execute “Generate $(-)$-RLITT” input T_{r-} and T_{s-}
    2. Set T_{r+} to be the set of $(+)$-RLITT with TCN equal to 0
    3. Set T_{r-} to be the set of $(-)$-RLITT with TCN equal to 0
    4. Set T_{s+} to be the set of $(+)$-RLITT good scions with TCN equal to i
    5. Set T_{s-} to be the set of $(-)$-RLITT good scions with TCN equal to i
    6. Execute “Generate $(+)$-RLITT” input T_{r+} and T_{s+}
    7. Execute “Generate $(-)$-RLITT” input T_{r-} and T_{s-}
    8. Add the results to T

TECHNOLOGIES

FUTURE WORK

MINIMALIZATION

OF ARBORESCENT REPRESENTATIVES



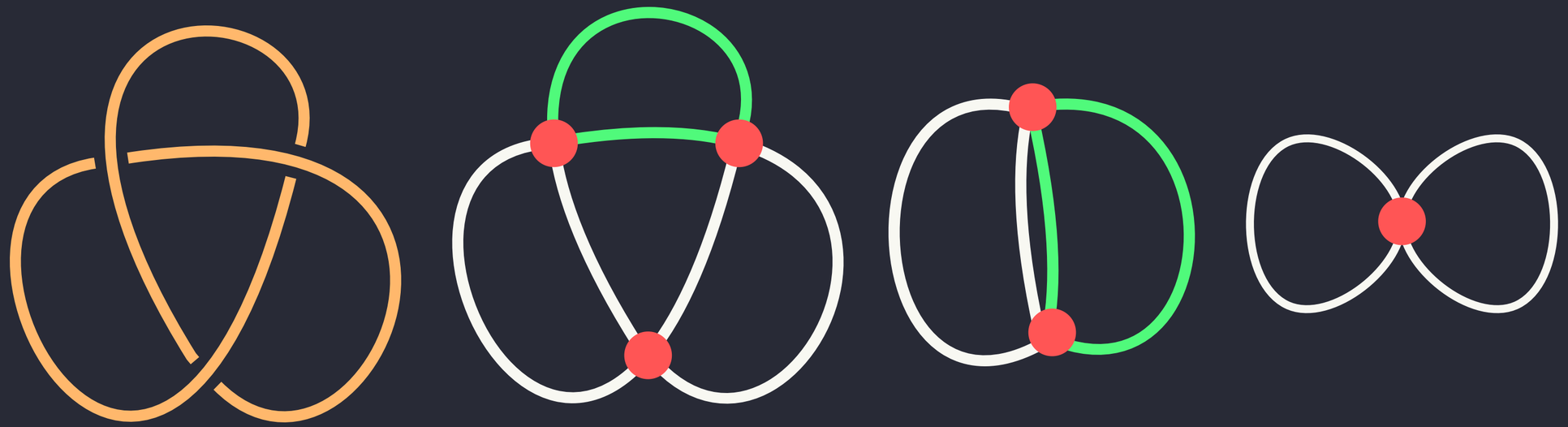
MINIMALIZATION

GENERAL REPRESENTATIVES

TABULATION OF POLYGONAL TANGLES

ALGEBRAIC

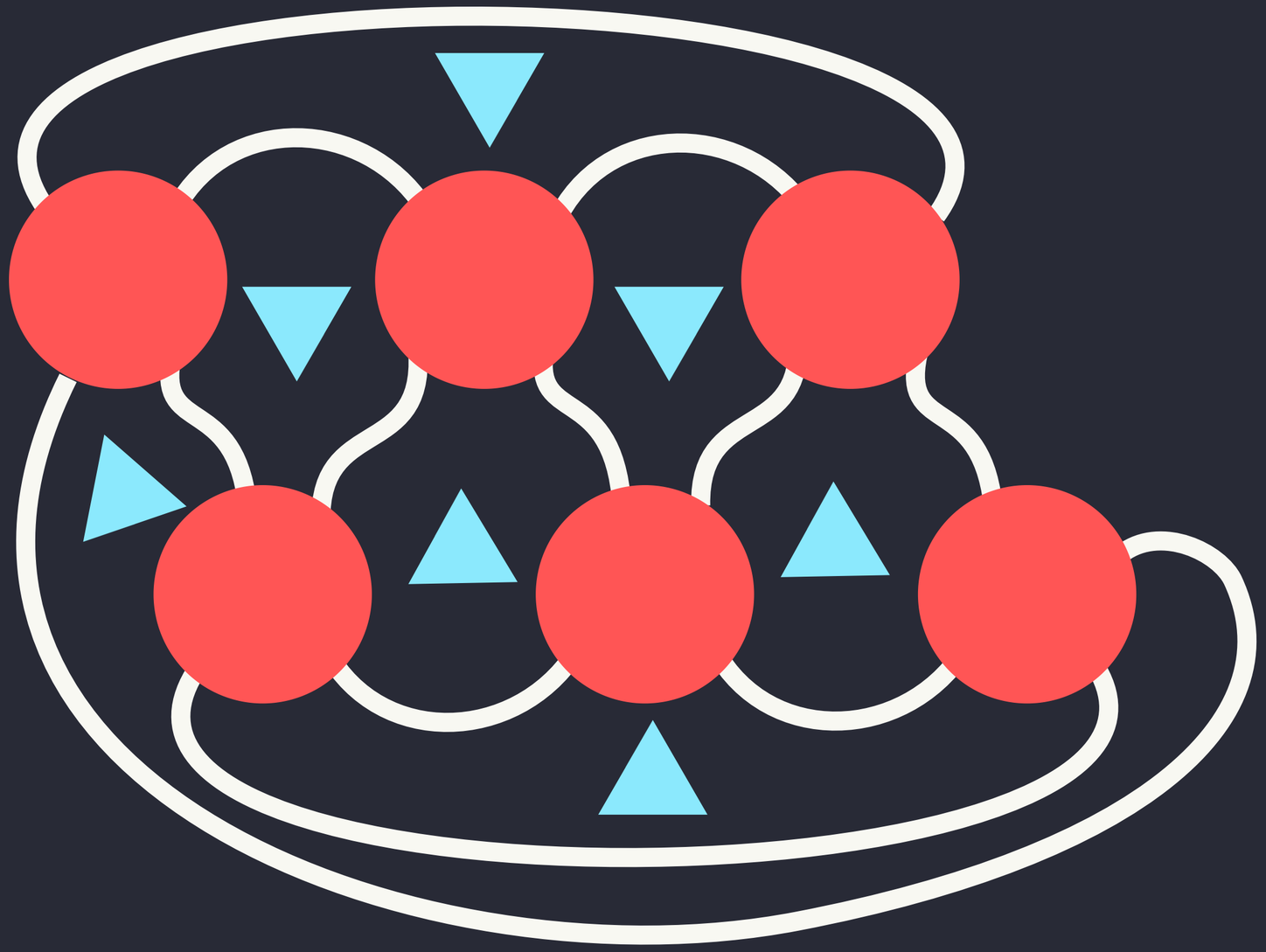
$+$, $\vee$, and plumbing only form bigons between basic tangles in the “knot shadow”.

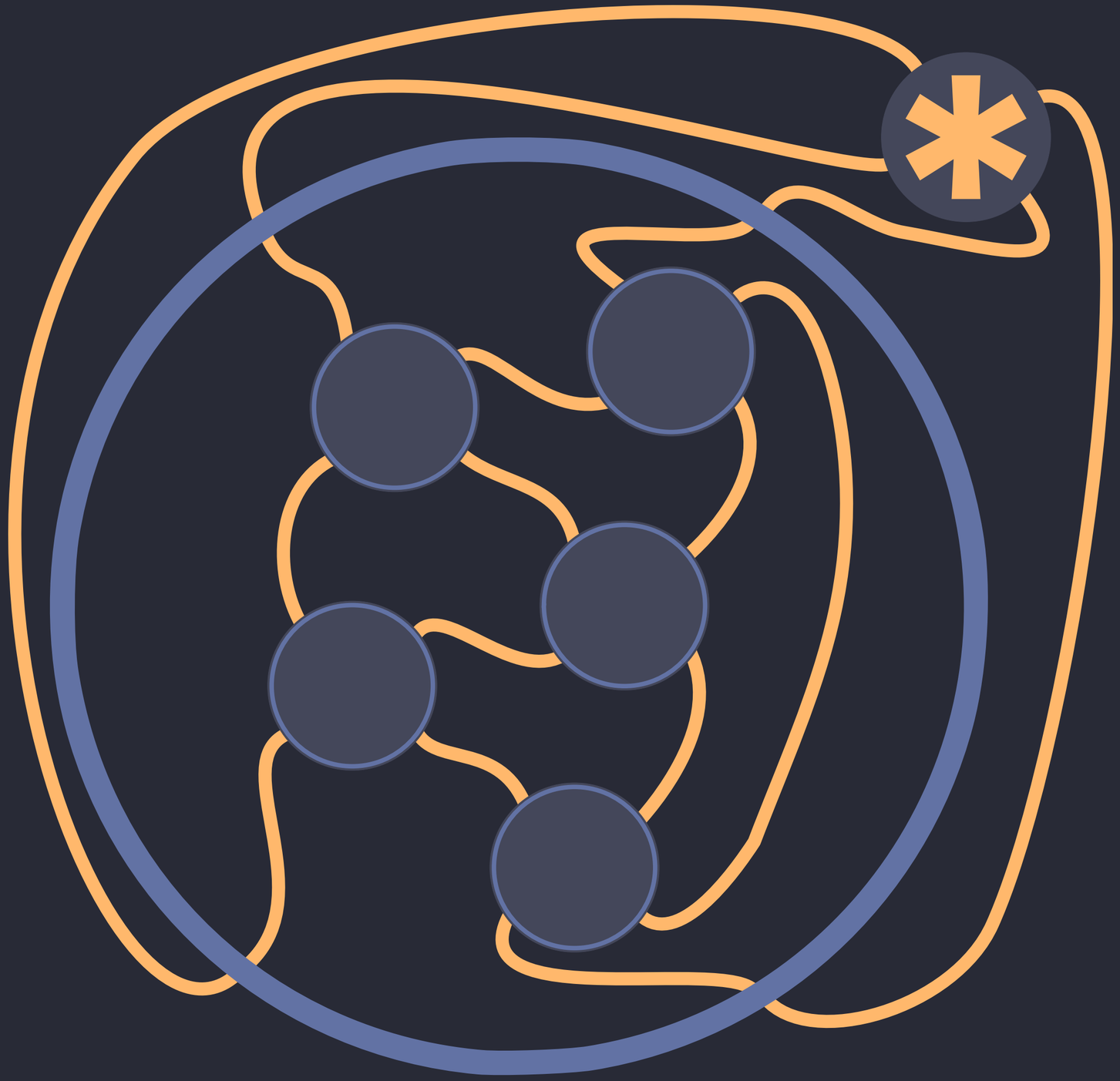


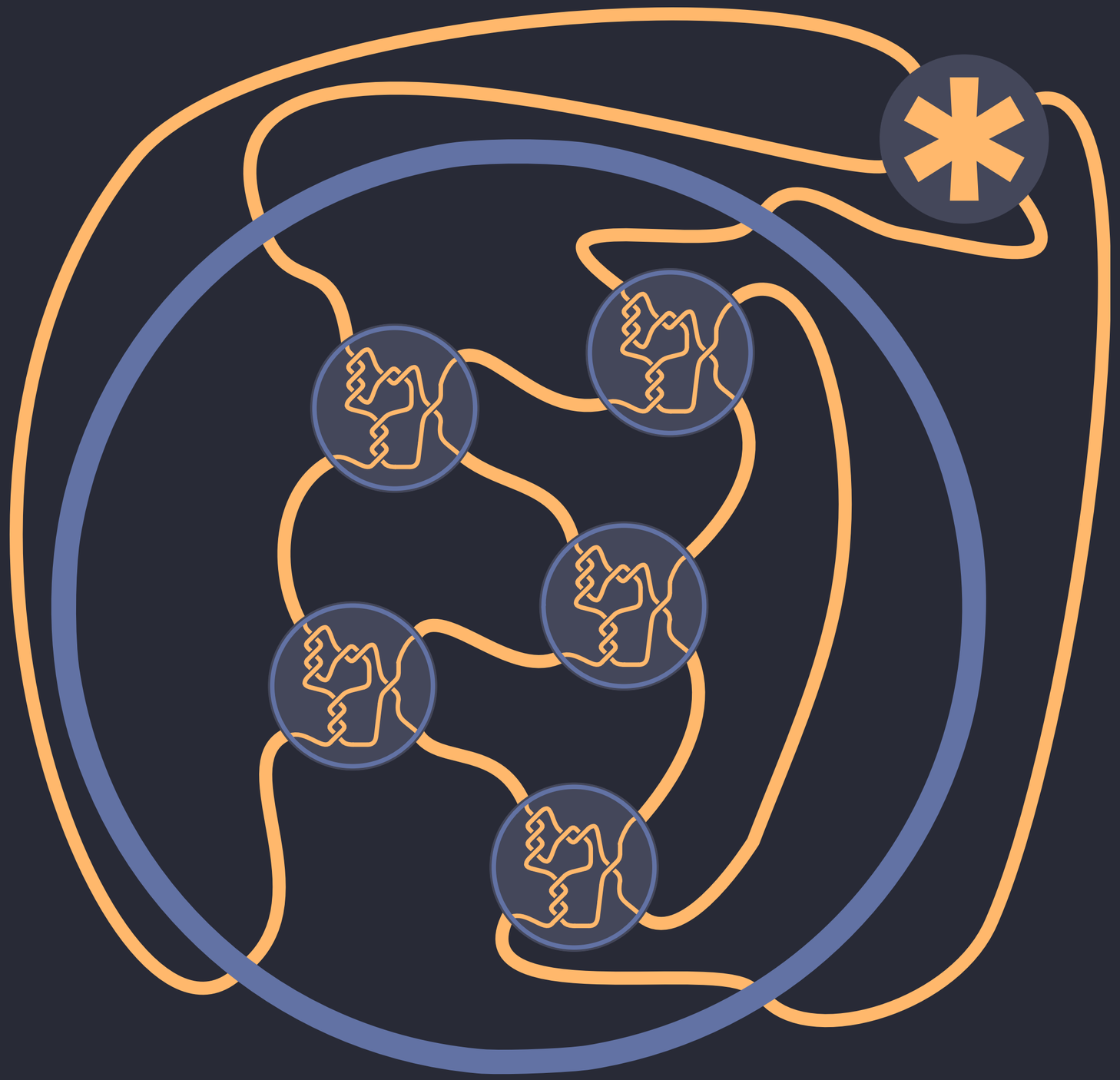
POLYGONAL TANGLES

HOW TO FORM SOMETHING OTHER THAN BIGONS?

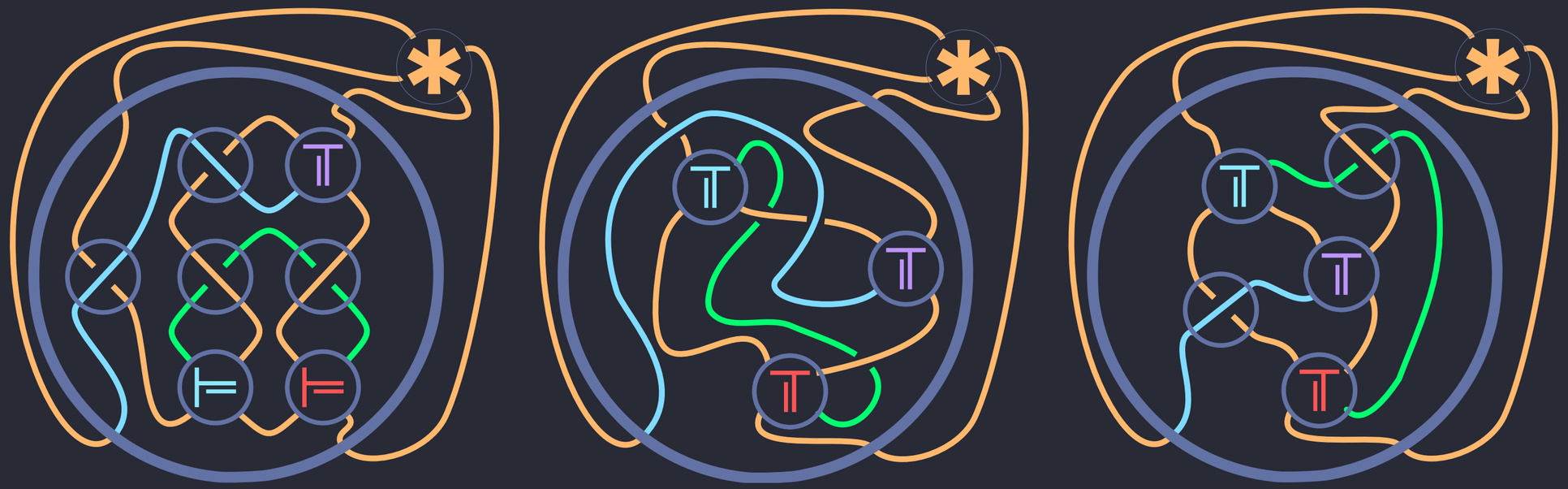
6**







MINIMALLY POLYGONAL ARBORESCENT TANGLES



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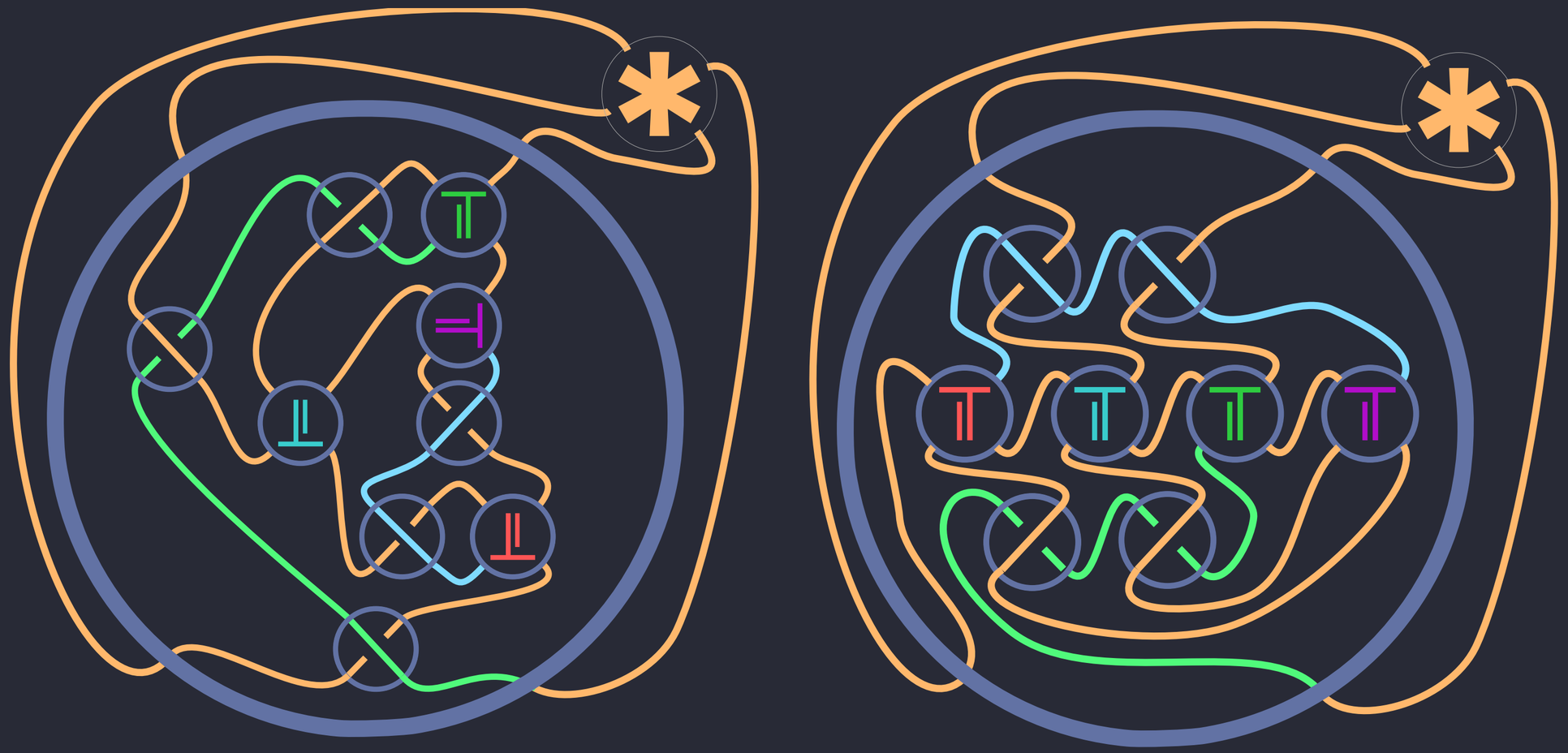

SOLUTION?

TABULATE THE POLYGONAL TANGLES.

THANK YOU!

SPECIAL THANKS TO THE COMMITTEE MEMBERS:

- Dr. Isabel Darcy, Supervisor
- Dr. Francis Bonahon
- Dr. Keiko Kawamuro
- Dr. Colleen Mitchell
- Dr. Radmila Sazdanović

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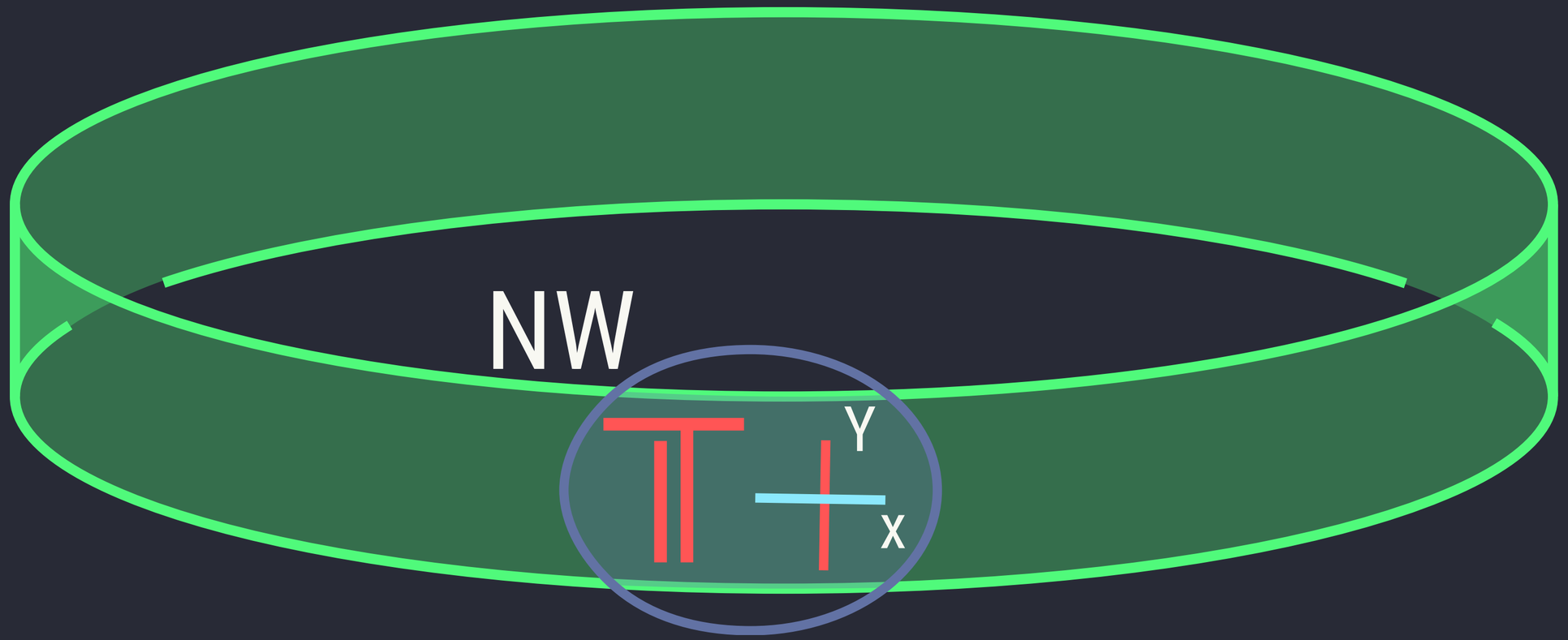
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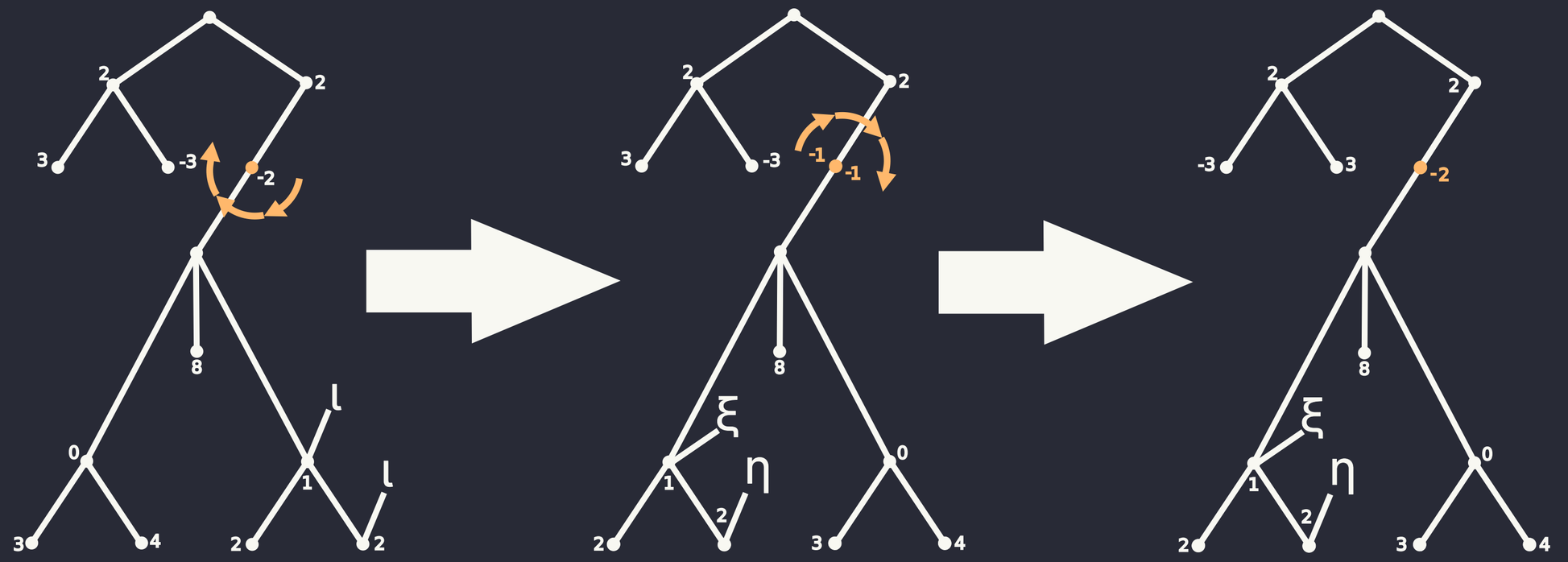
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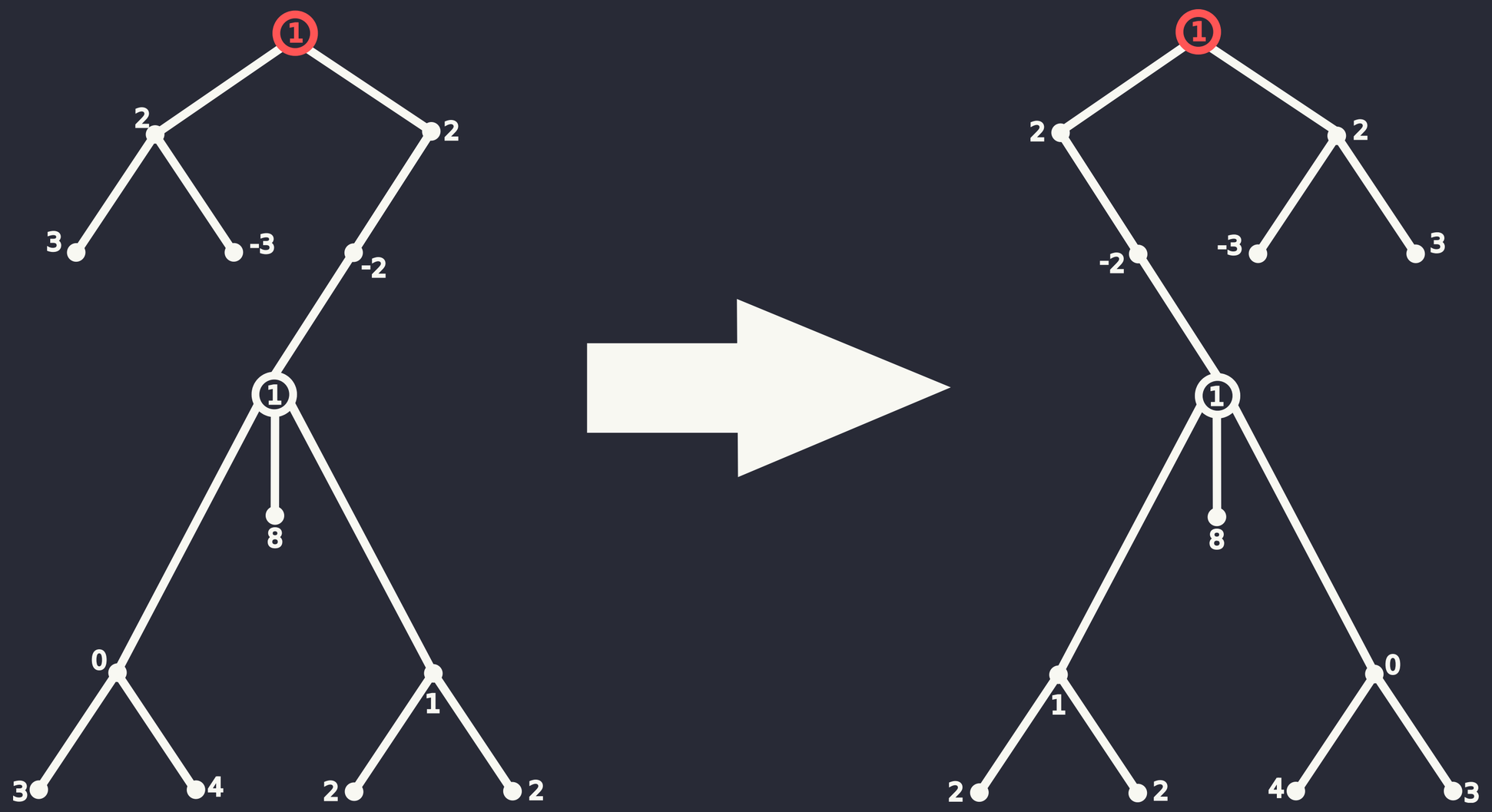
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